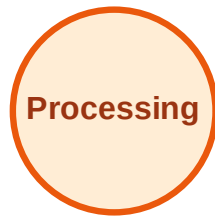


BFS Traversal

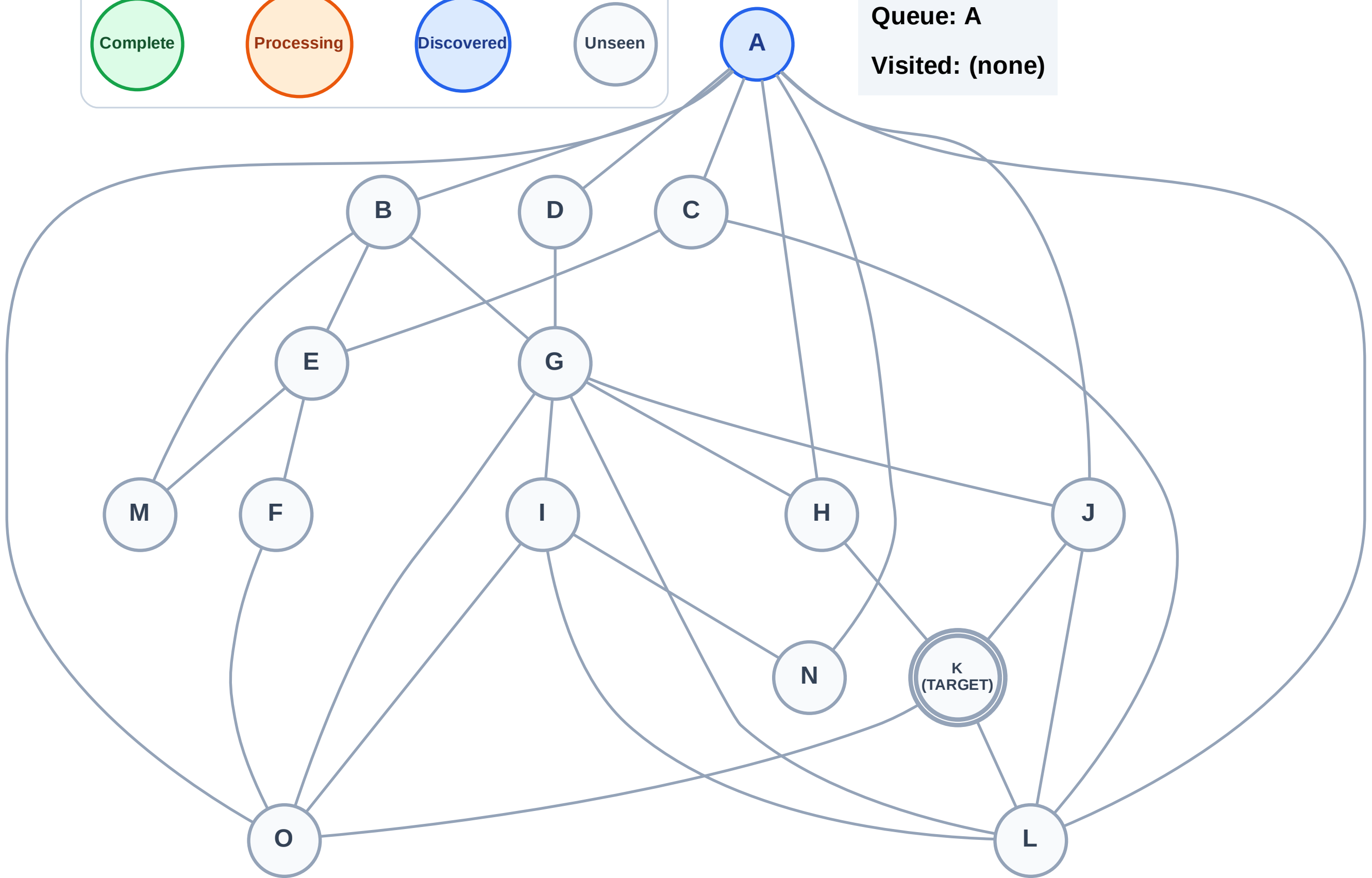
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



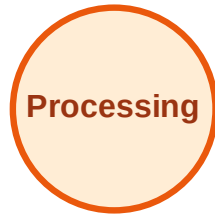
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



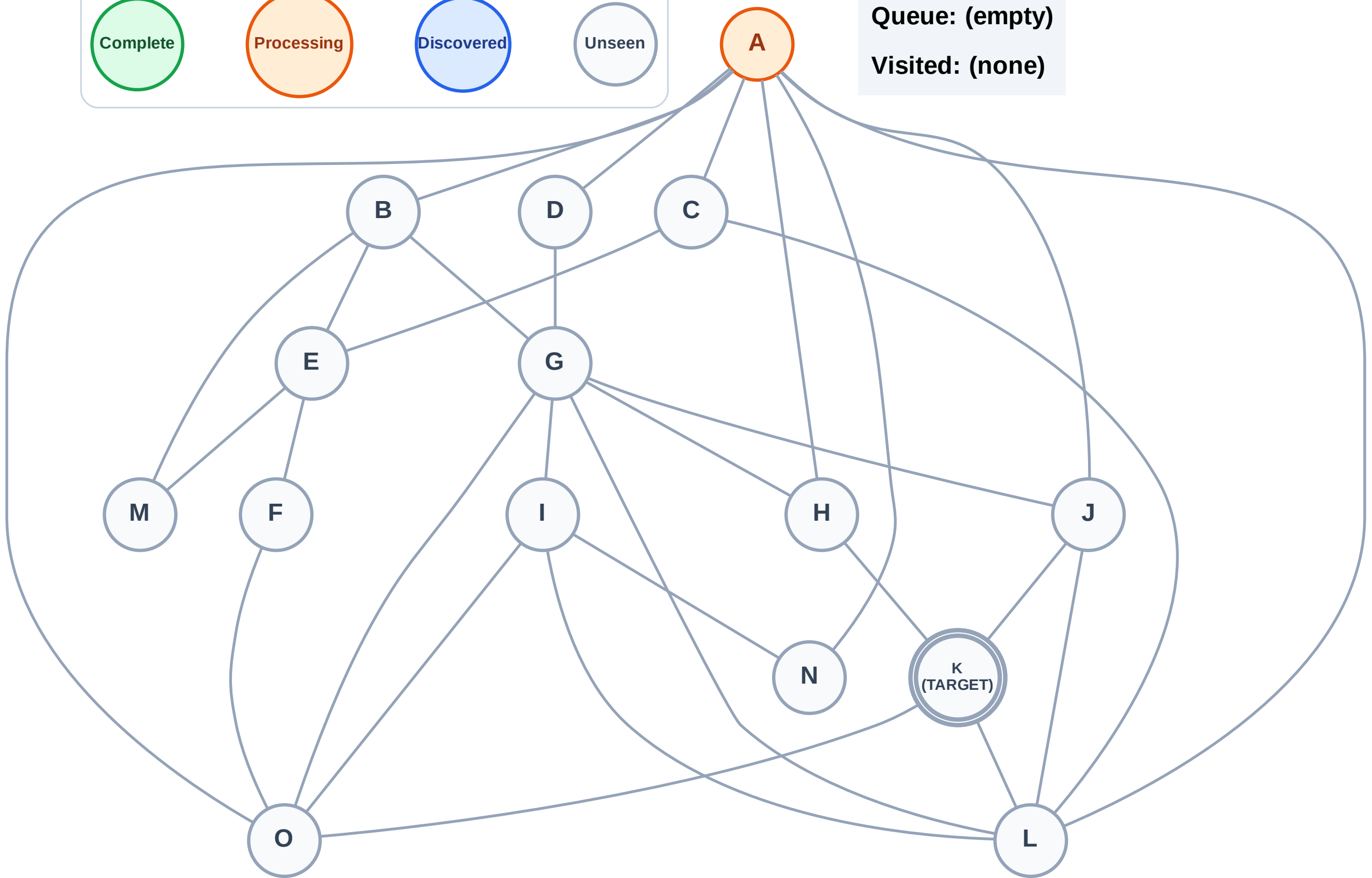
Discovered



Unseen

Queue: (empty)

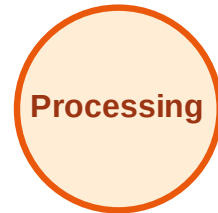
Visited: (none)



BFS Traversal

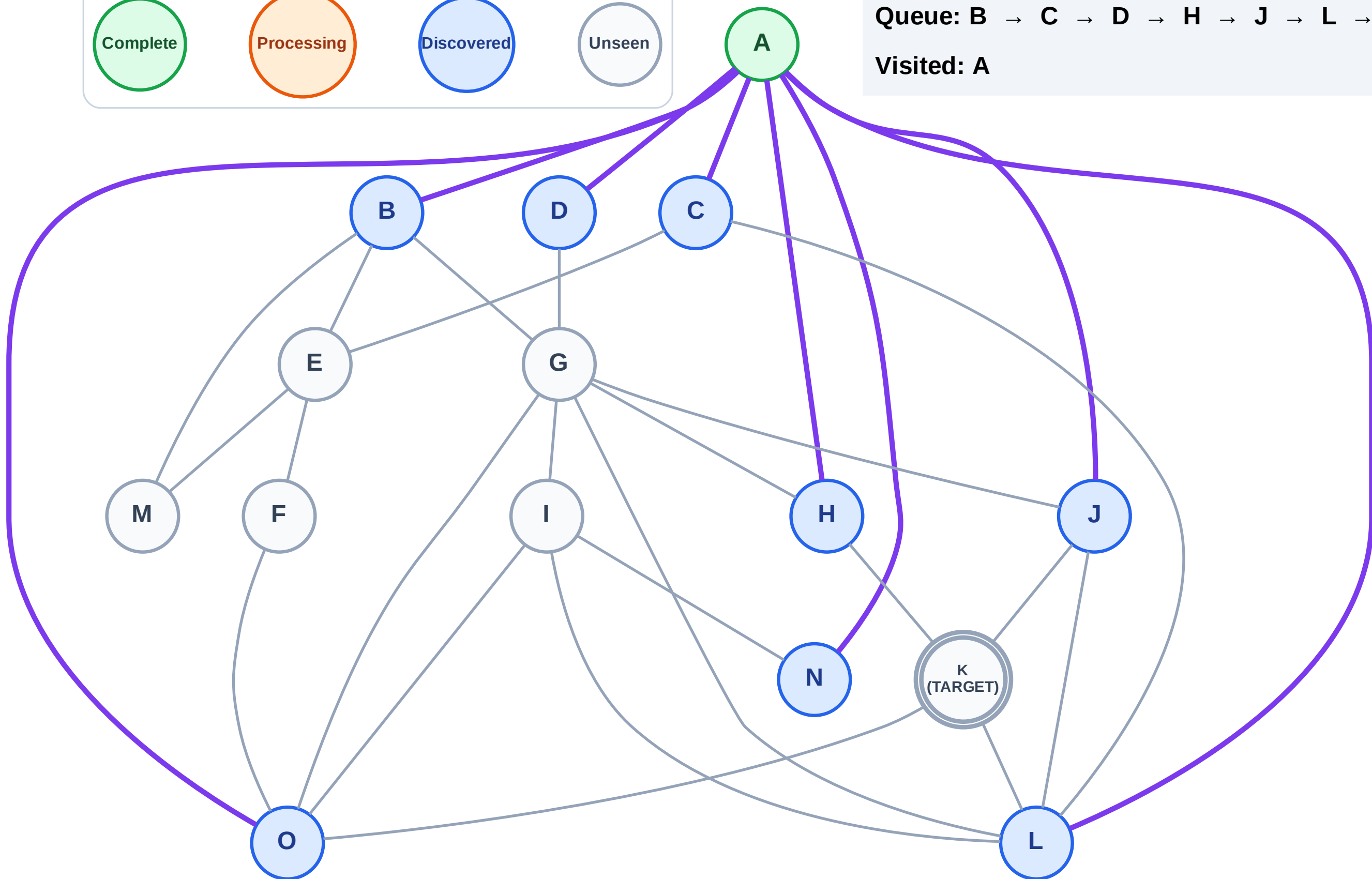
Step 3: Discover B, C, D, H, J, L, N, O from A

Legend



Queue: B → C → D → H → J → L → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

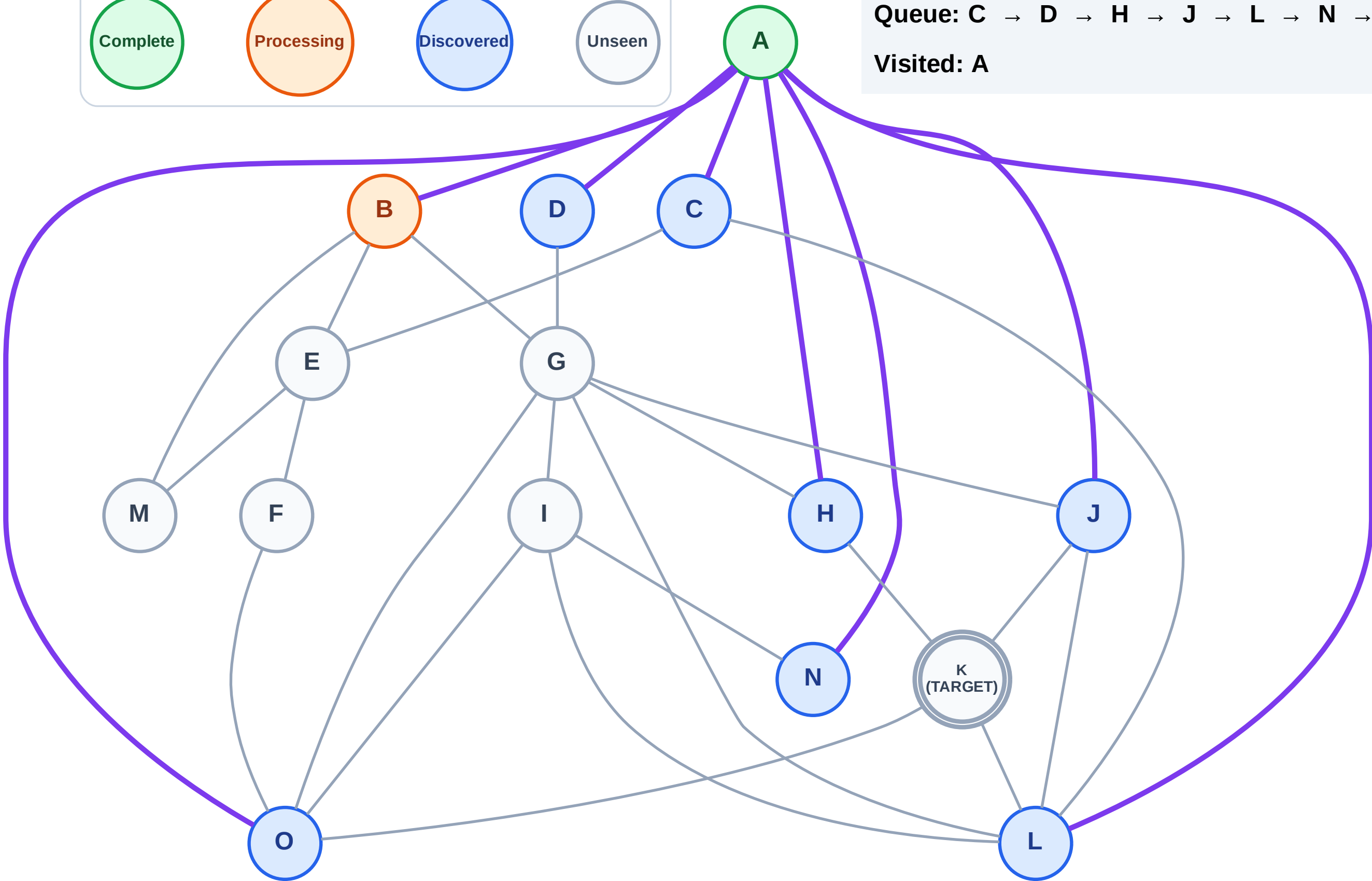
Processing

Discovered

Unseen

Queue: C → D → H → J → L → N → O

Visited: A



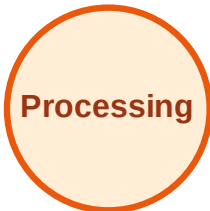
BFS Traversal

Step 5: Discover E, G, M from B

Legend



Complete



Processing



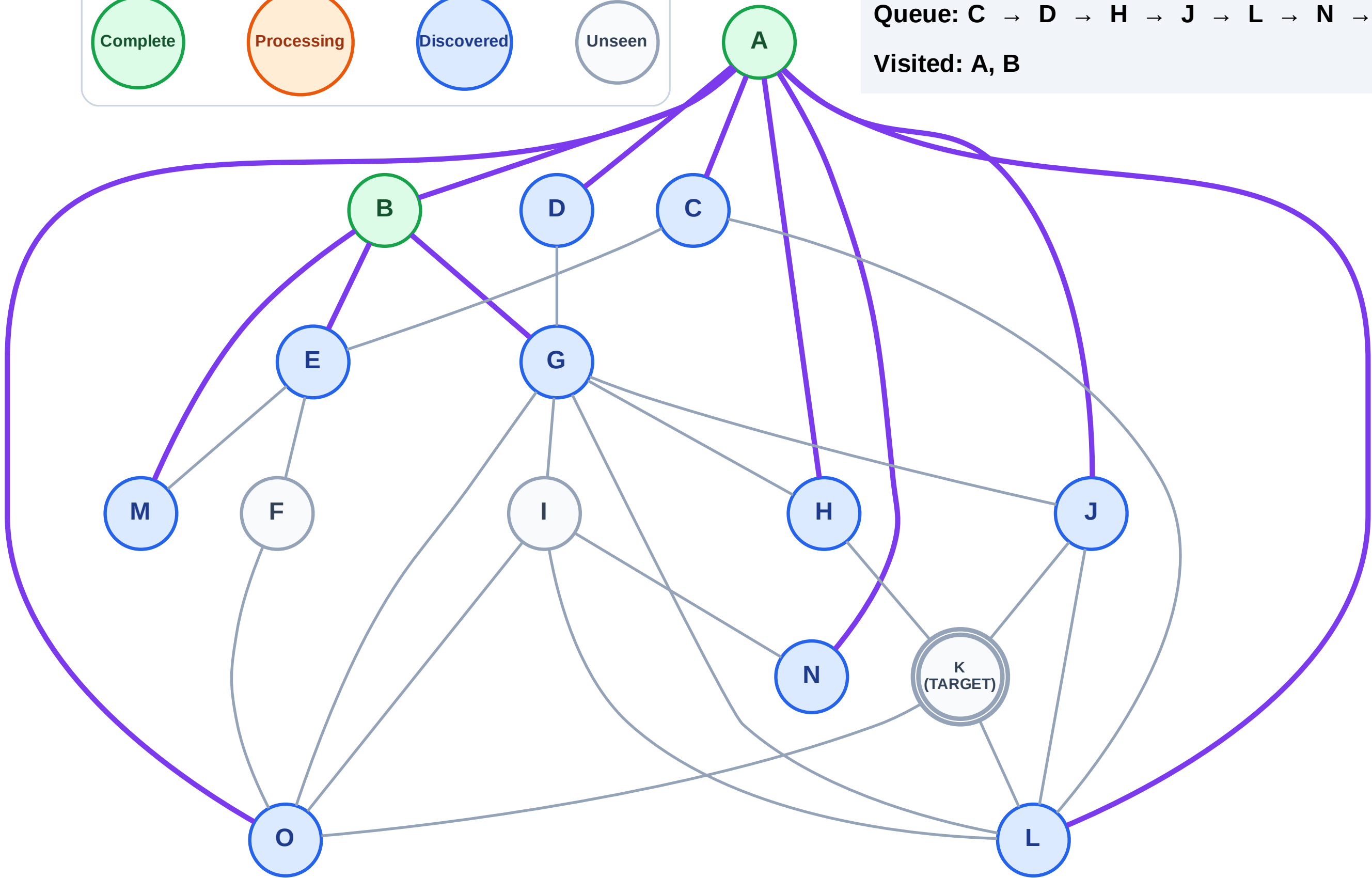
Discovered



Unseen

Queue: C → D → H → J → L → N → O → E → G → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

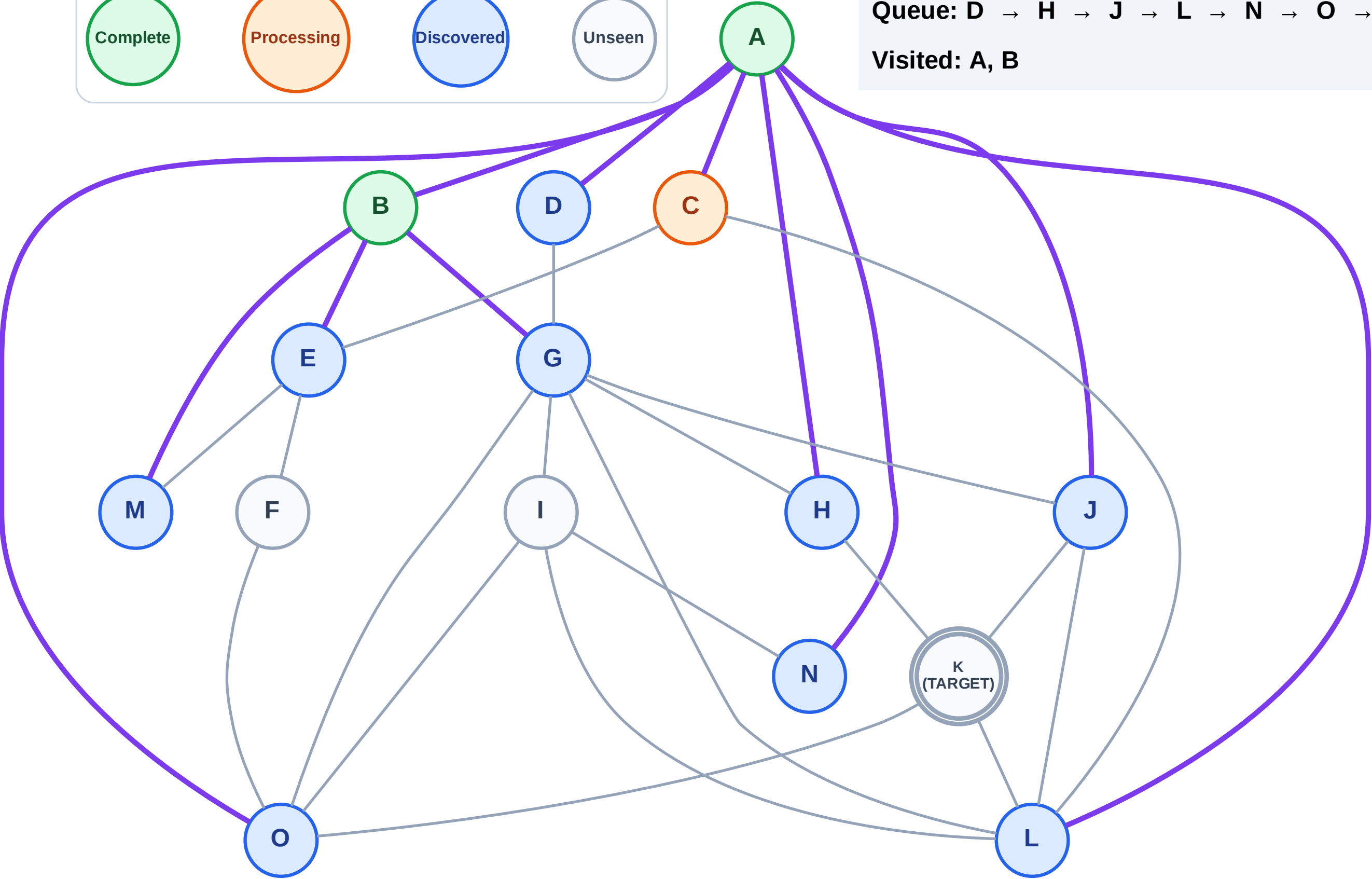
Processing

Discovered

Unseen

Queue: D → H → J → L → N → O → E → G → M

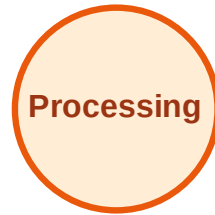
Visited: A, B



BFS Traversal

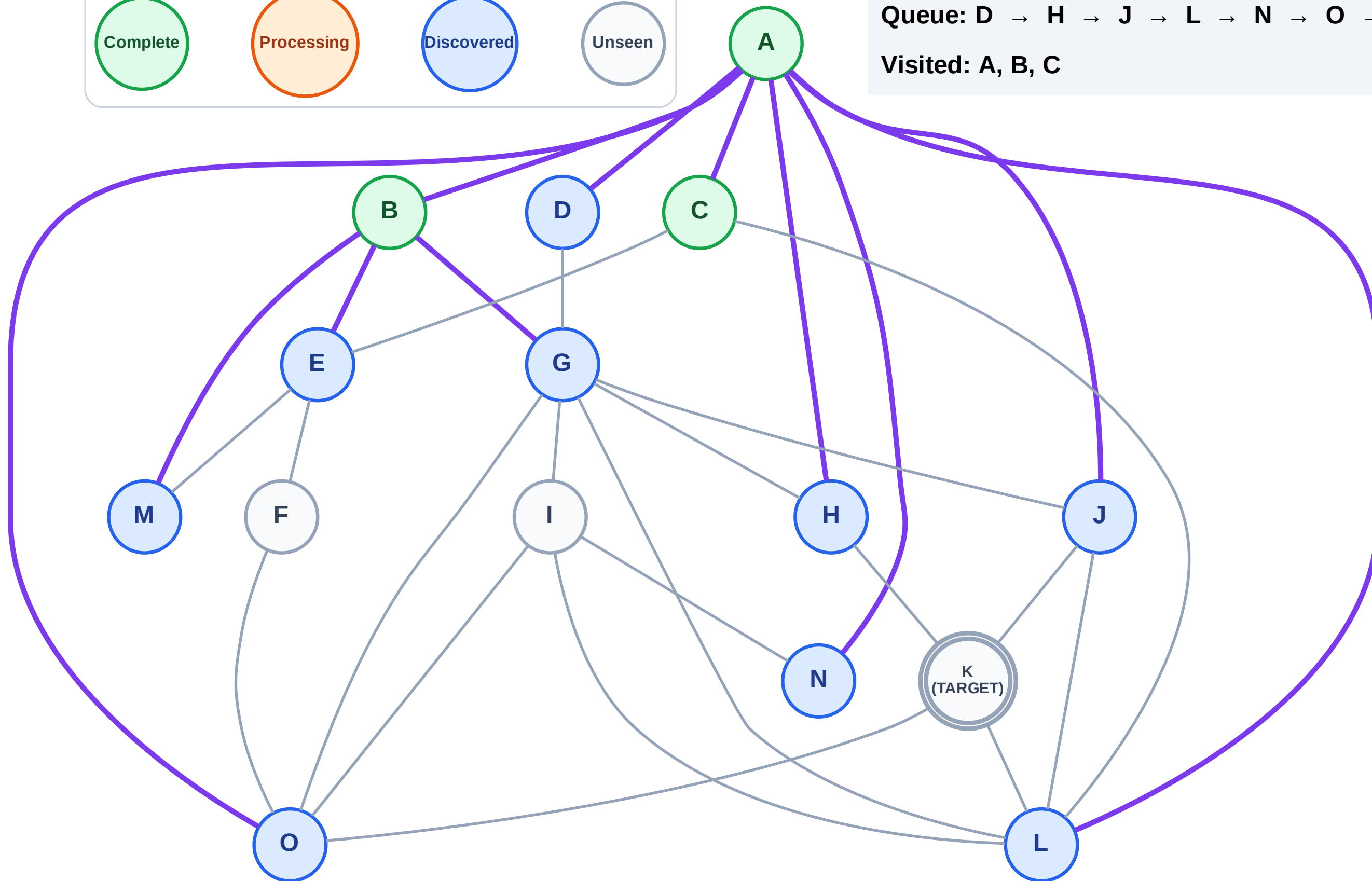
Step 7: No new neighbors from C

Legend



Queue: D → H → J → L → N → O → E → G → M

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

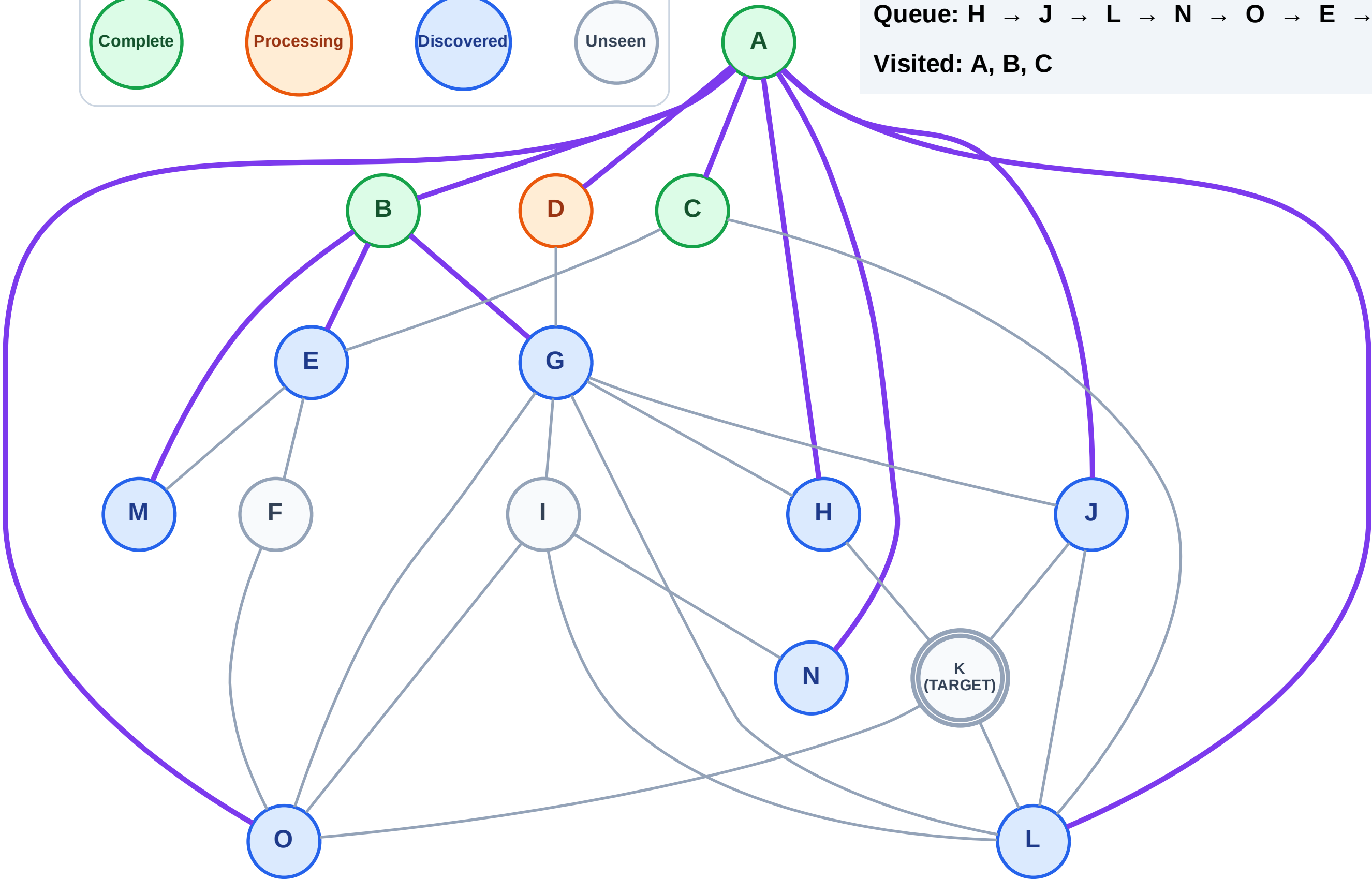
Processing

Discovered

Unseen

Queue: H → J → L → N → O → E → G → M

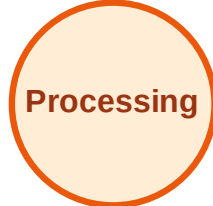
Visited: A, B, C



BFS Traversal

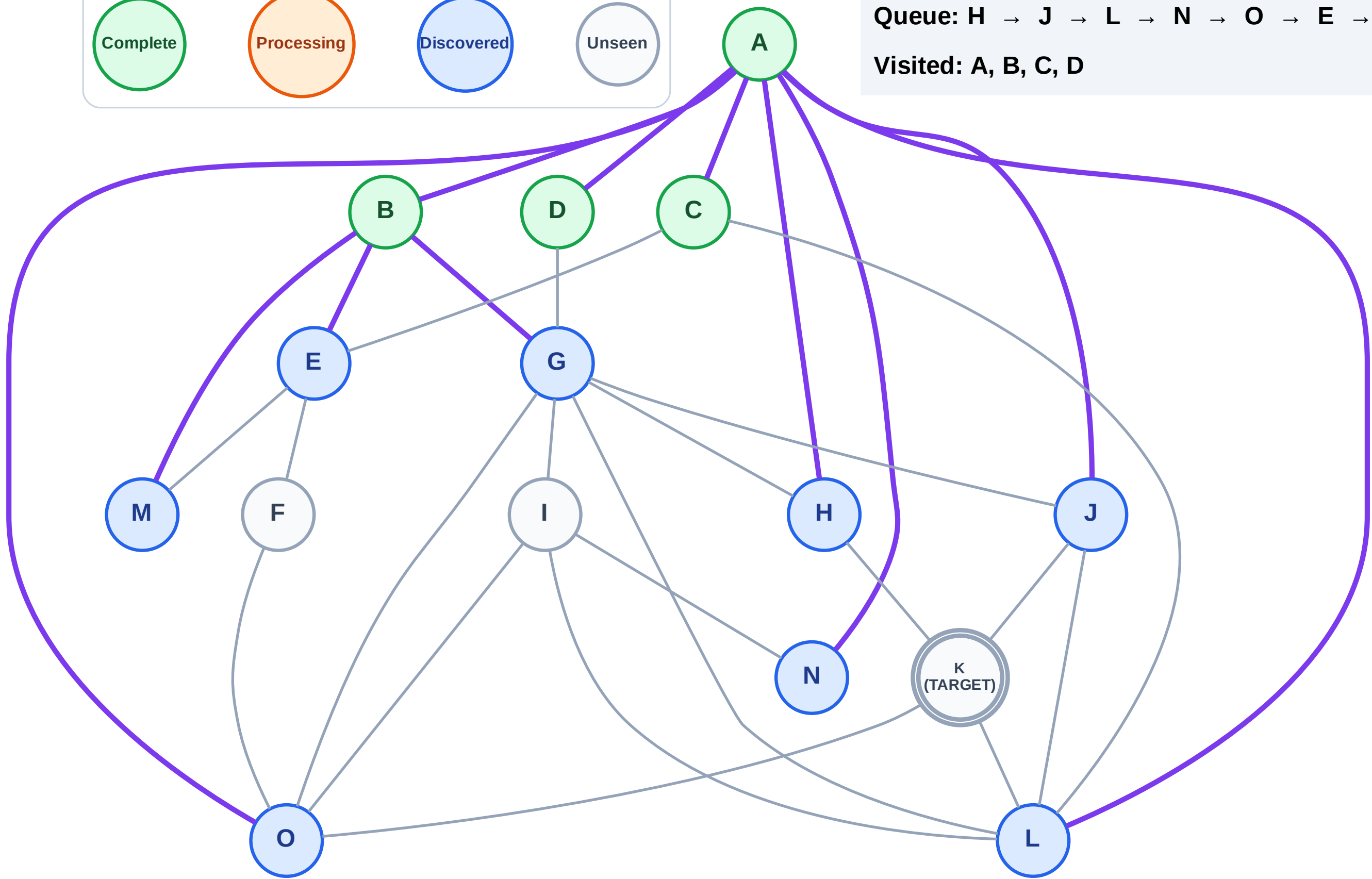
Step 9: No new neighbors from D

Legend



Queue: H → J → L → N → O → E → G → M

Visited: A, B, C, D



BFS Traversal

Step 10: Process node H

Legend

Complete

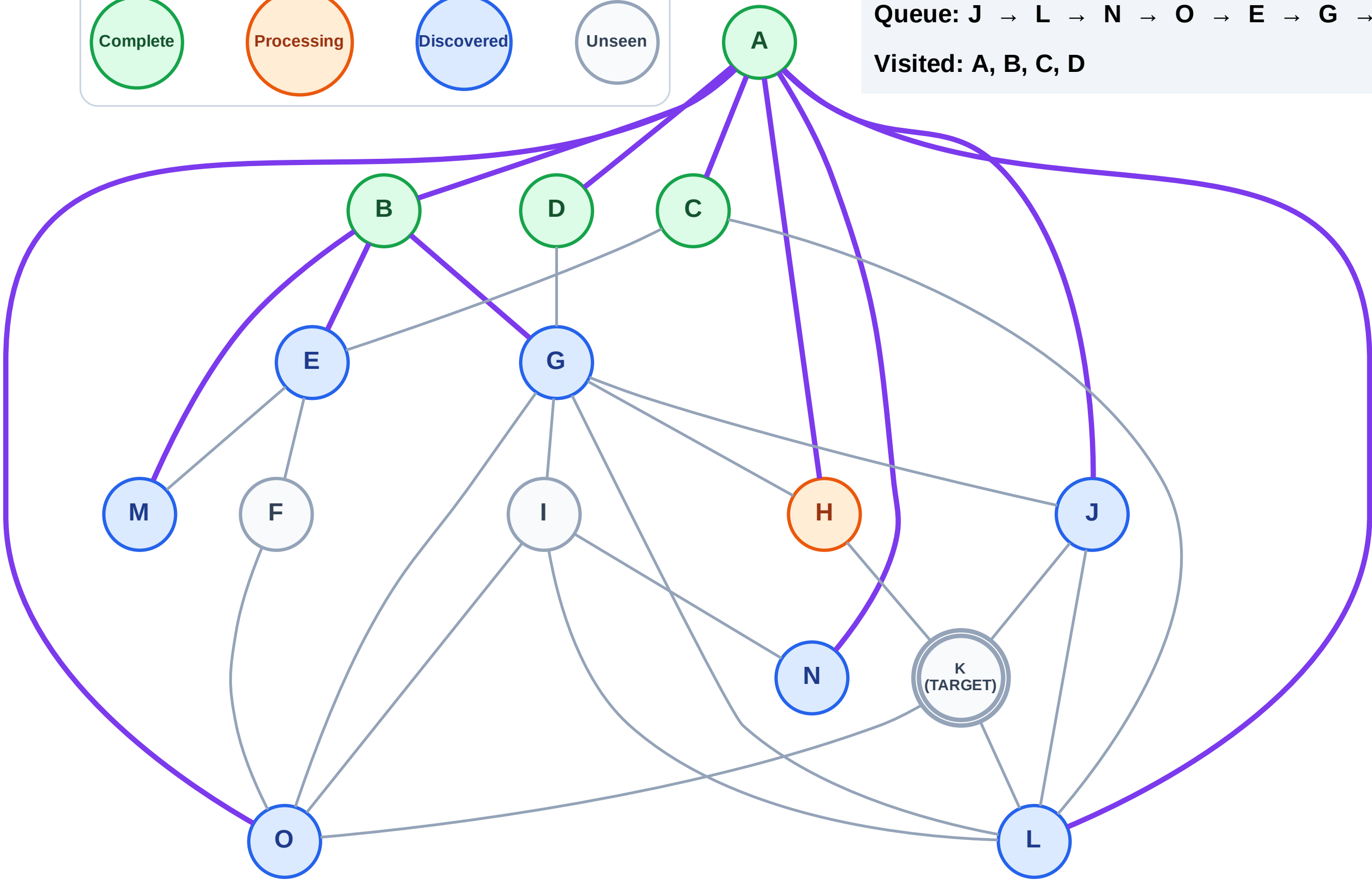
Processing

Discovered

Unseen

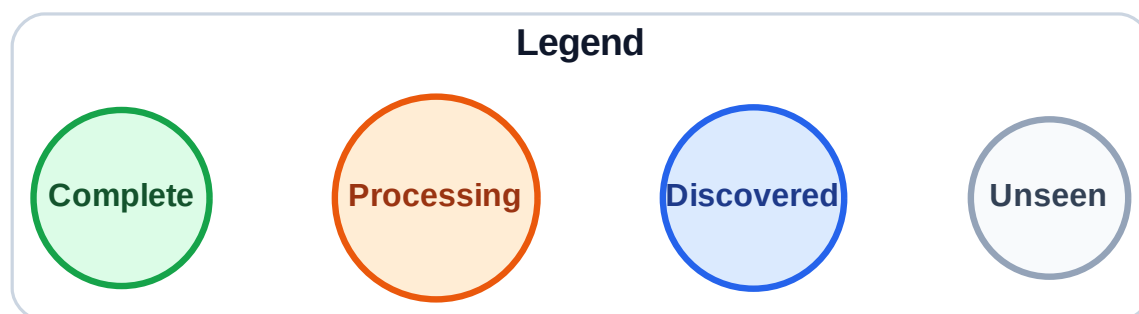
Queue: J → L → N → O → E → G → M

Visited: A, B, C, D



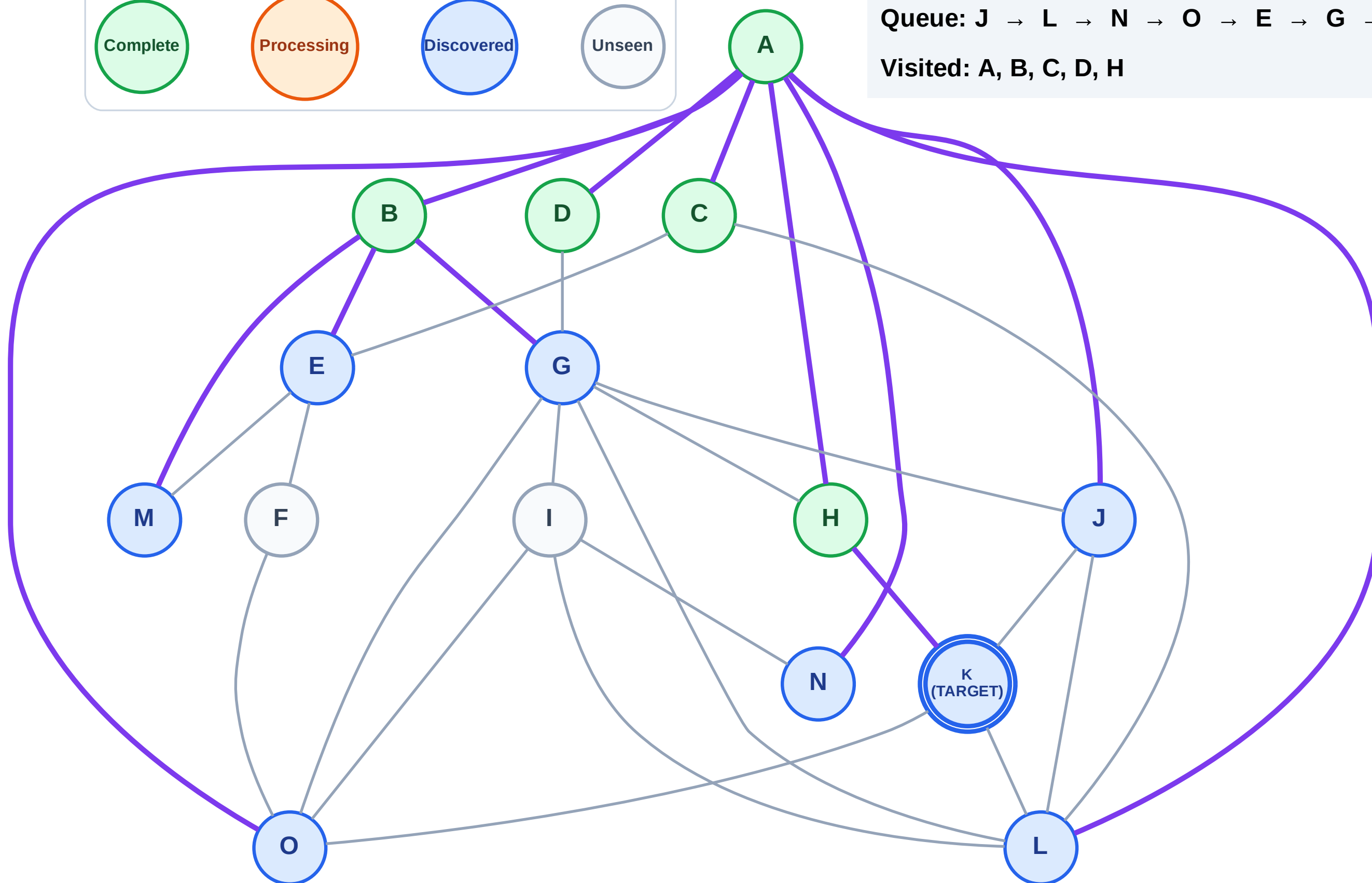
BFS Traversal

Step 11: Discover K from H



Queue: J → L → N → O → E → G → M → K

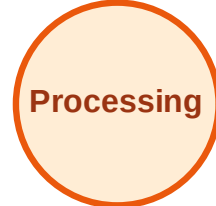
Visited: A, B, C, D, H



BFS Traversal

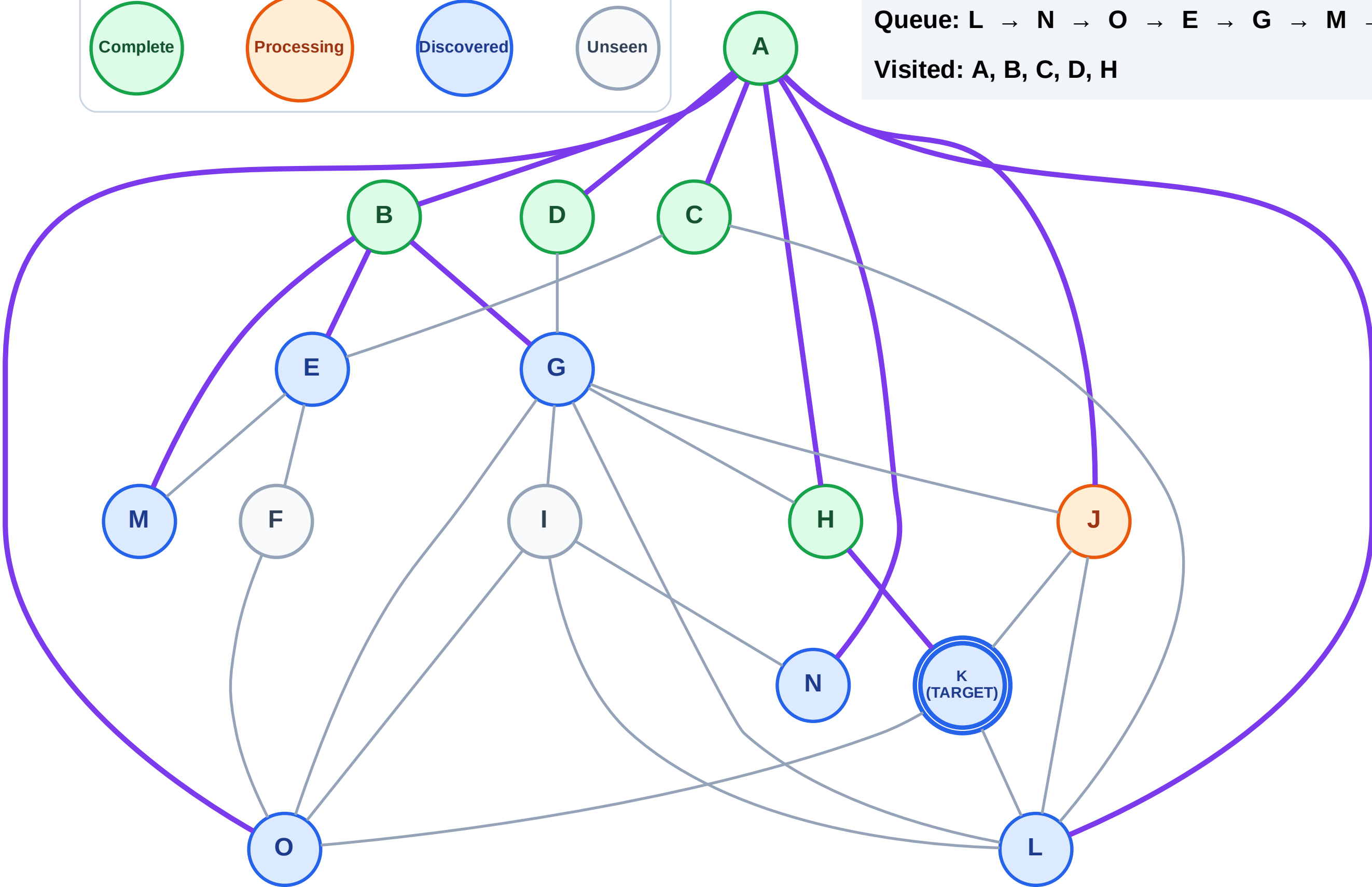
Step 12: Process node J

Legend



Queue: L → N → O → E → G → M → K

Visited: A, B, C, D, H



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

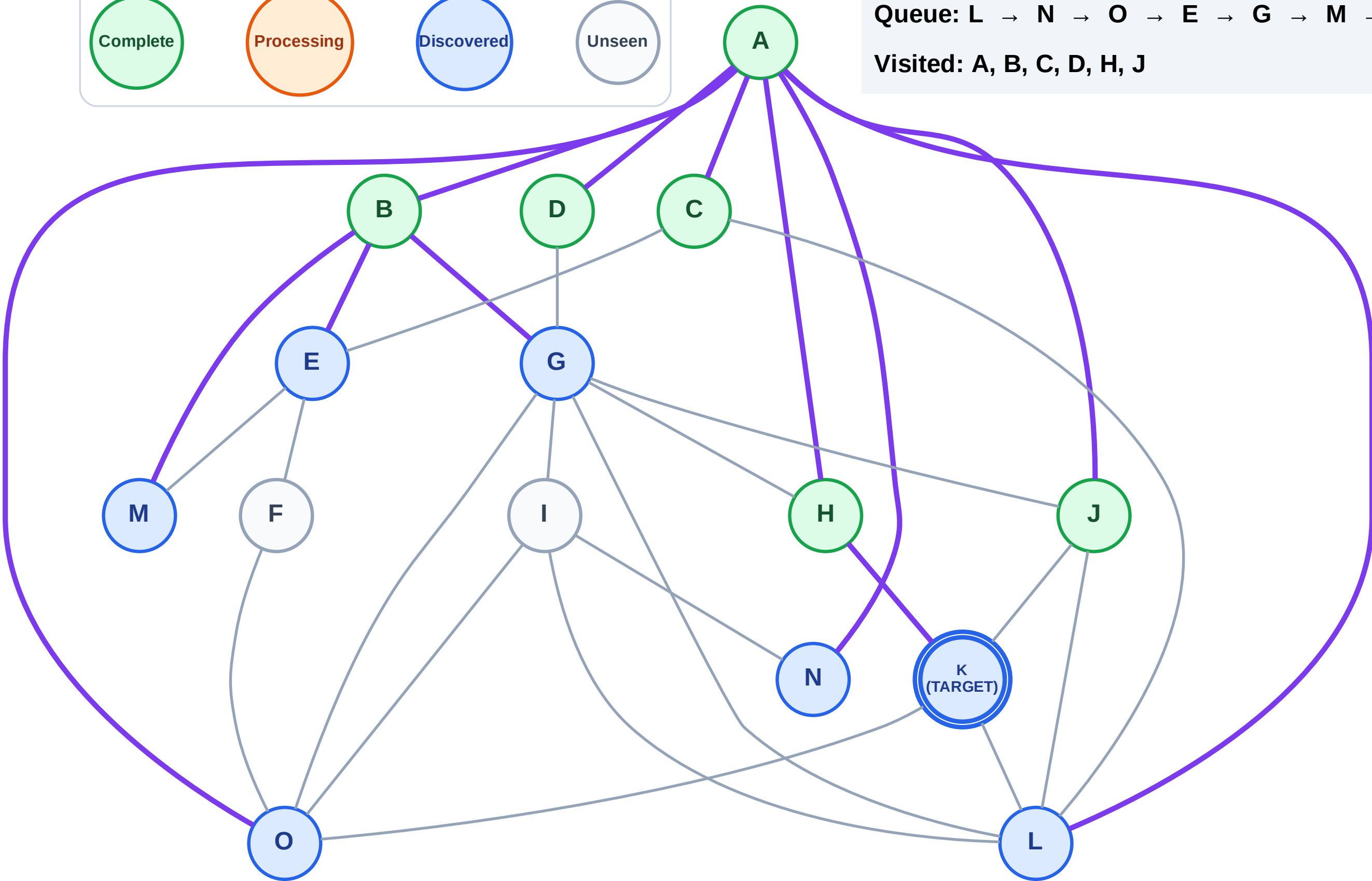
Processing

Discovered

Unseen

Queue: L → N → O → E → G → M → K

Visited: A, B, C, D, H, J



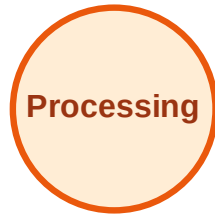
BFS Traversal

Step 14: Process node L

Legend



Complete



Processing



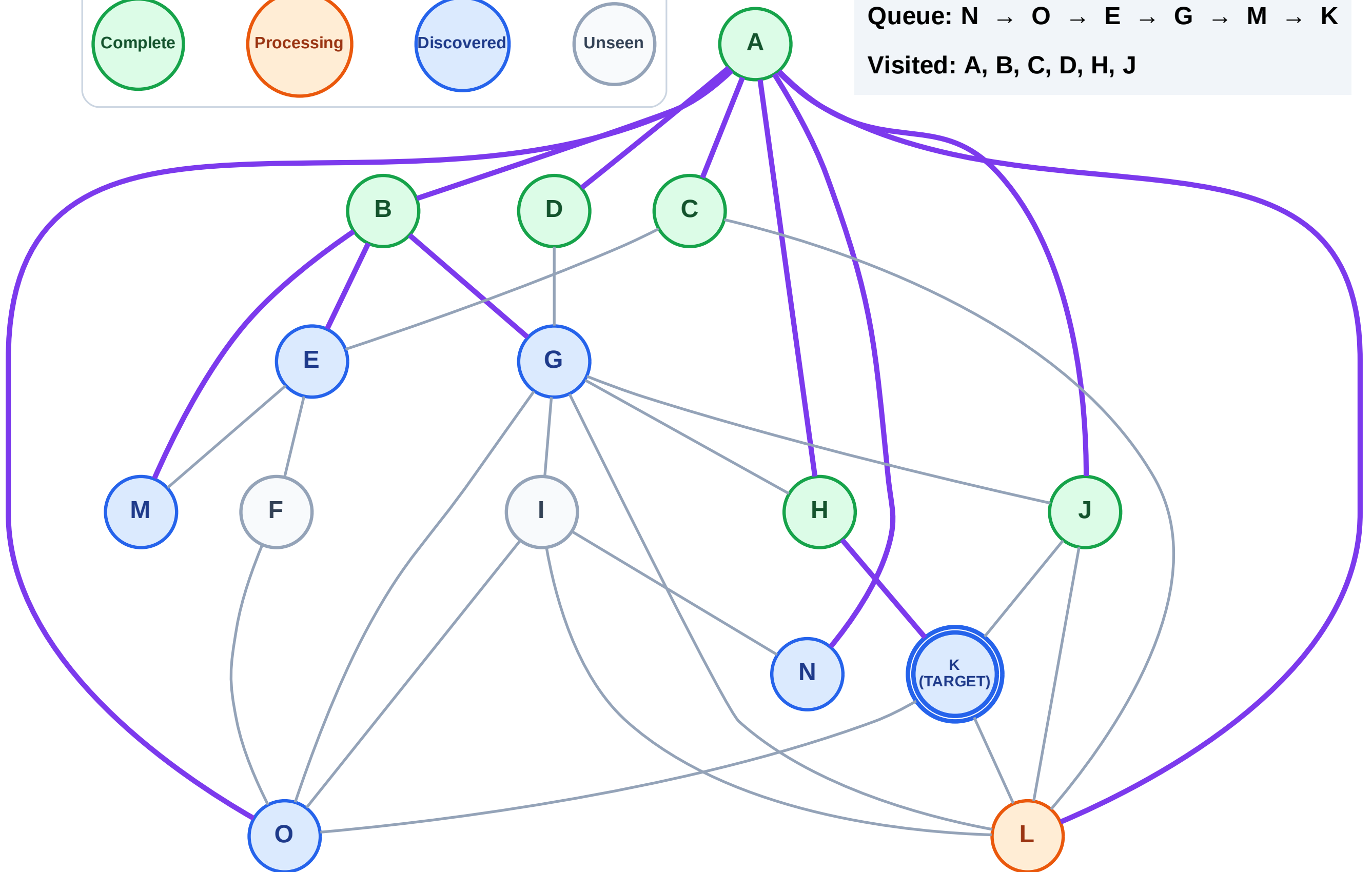
Discovered



Unseen

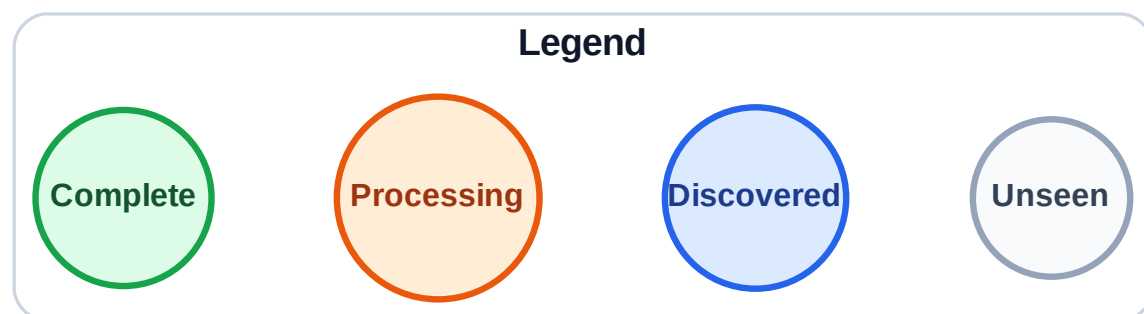
Queue: N → O → E → G → M → K

Visited: A, B, C, D, H, J



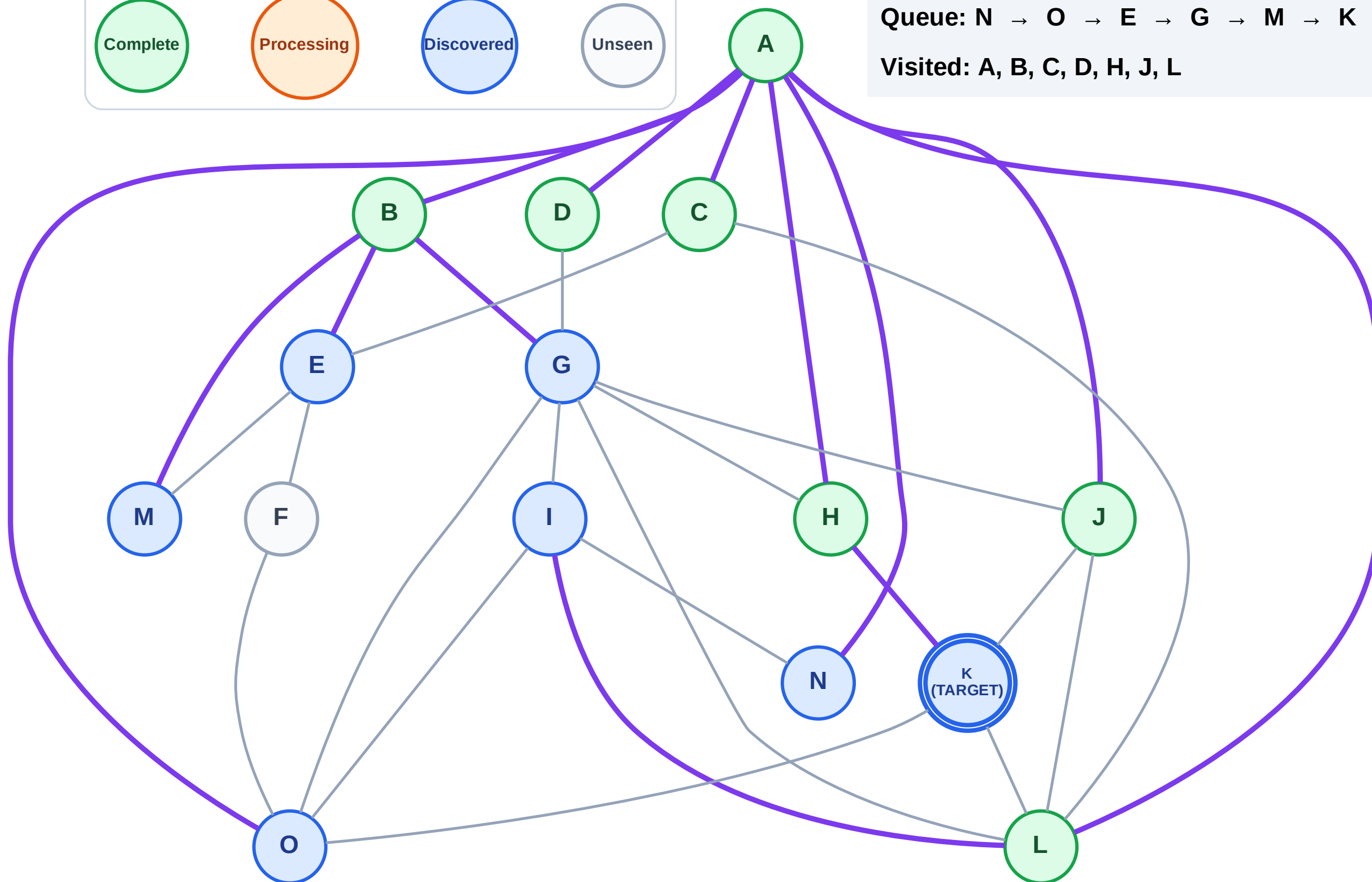
BFS Traversal

Step 15: Discover I from L



Queue: N → O → E → G → M → K → I

Visited: A, B, C, D, H, J, L



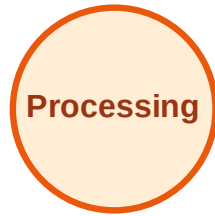
BFS Traversal

Step 16: Process node N

Legend



Complete



Processing



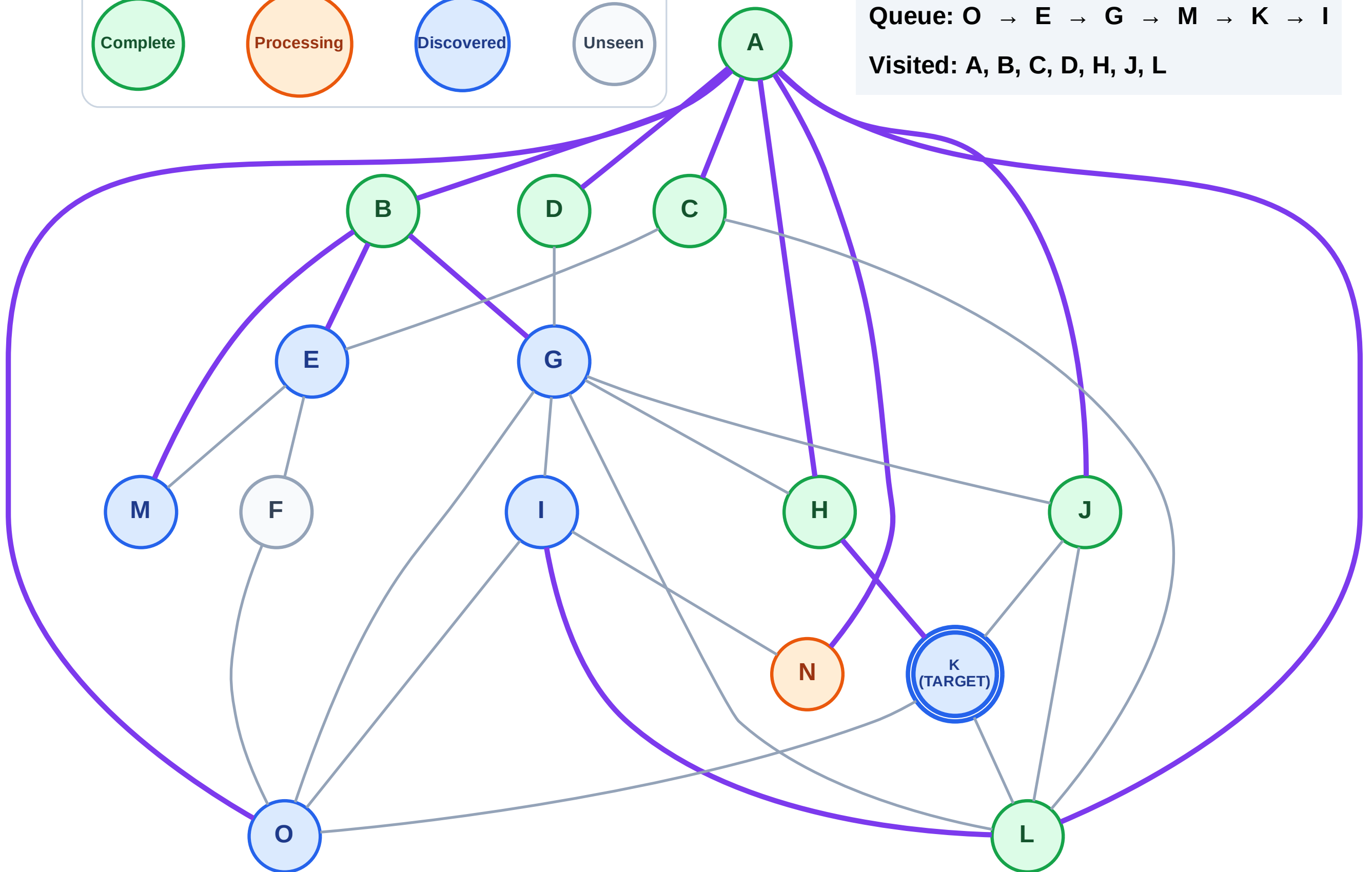
Discovered



Unseen

Queue: O → E → G → M → K → I

Visited: A, B, C, D, H, J, L



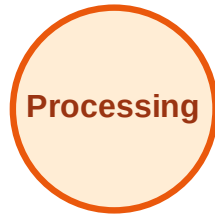
BFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



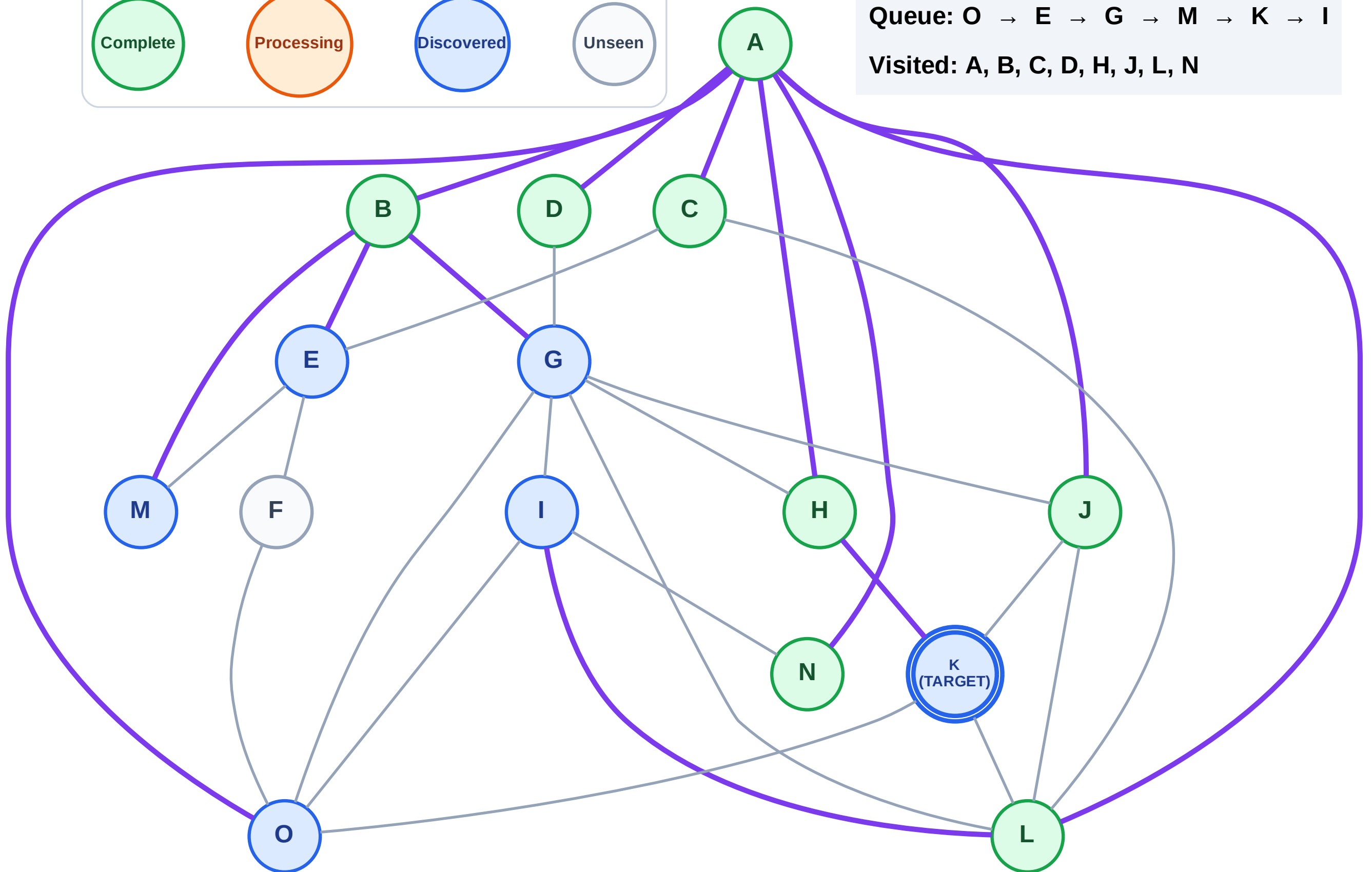
Discovered



Unseen

Queue: O → E → G → M → K → I

Visited: A, B, C, D, H, J, L, N



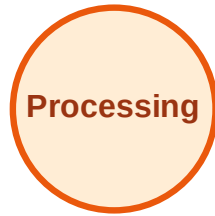
BFS Traversal

Step 18: Process node O

Legend



Complete



Processing



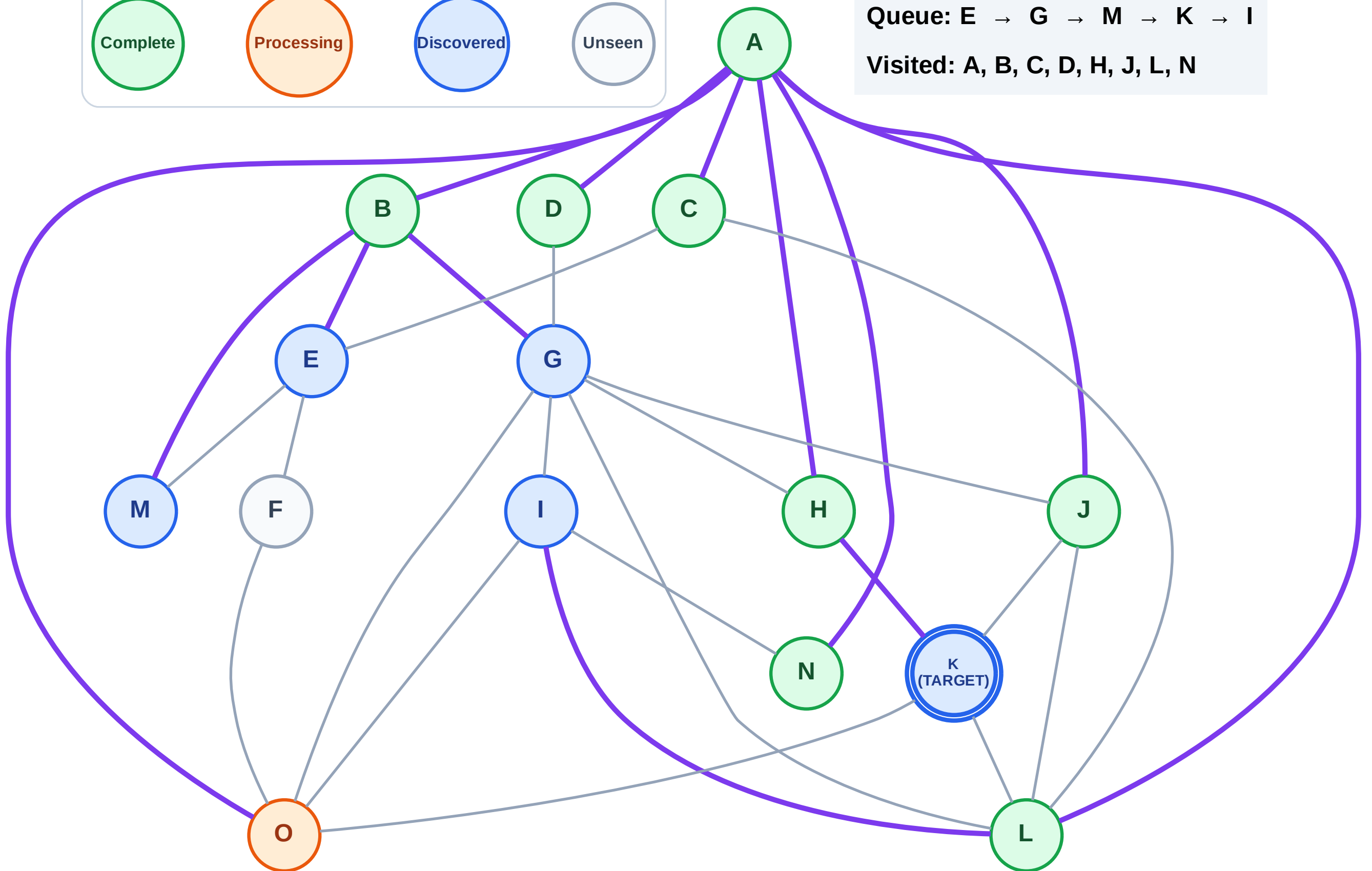
Discovered



Unseen

Queue: E → G → M → K → I

Visited: A, B, C, D, H, J, L, N



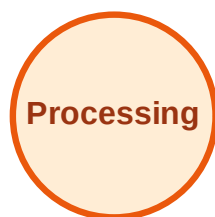
BFS Traversal

Step 19: Discover F from O

Legend



Complete



Processing



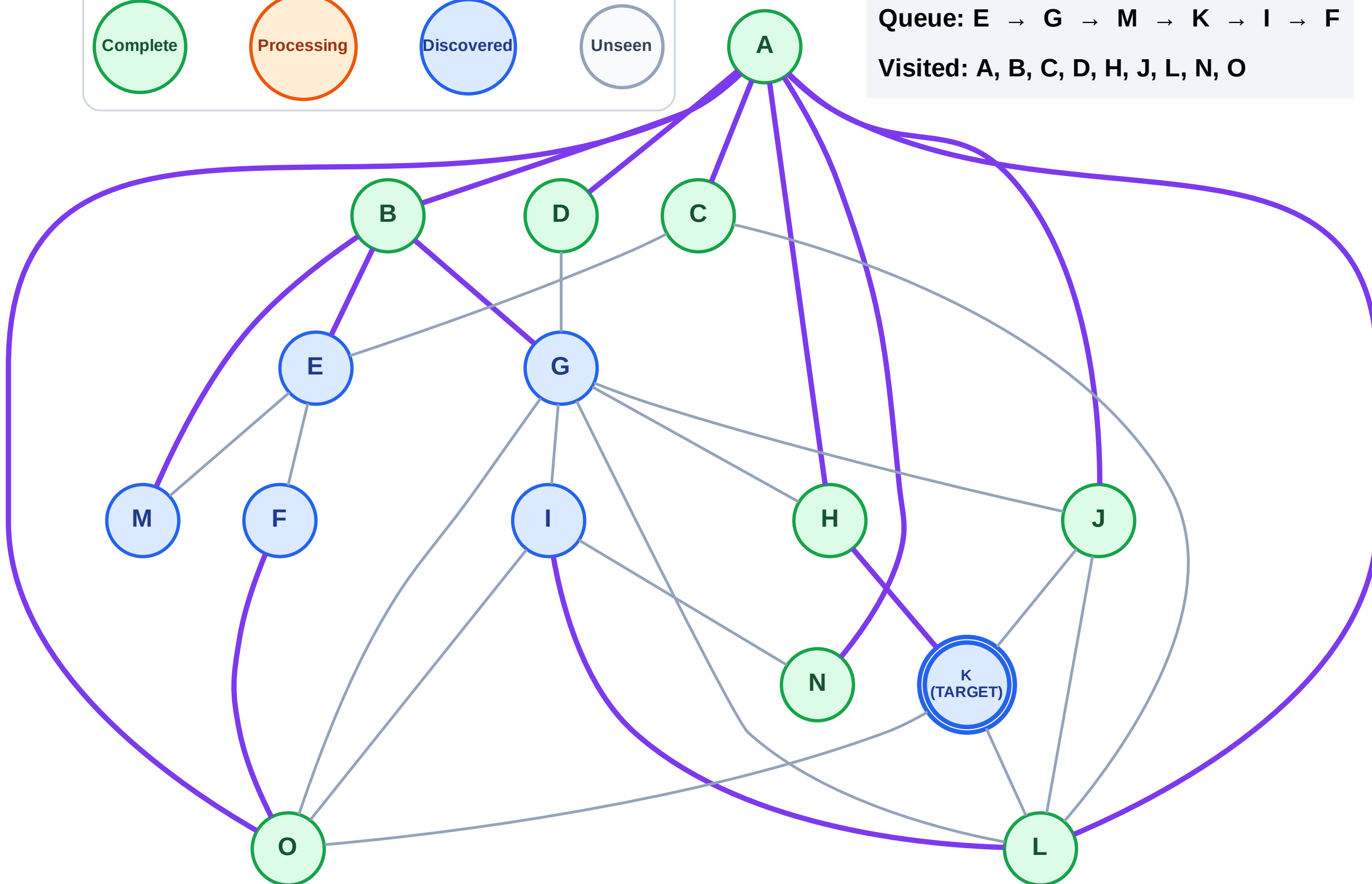
Discovered



Unseen

Queue: E → G → M → K → I → F

Visited: A, B, C, D, H, J, L, N, O



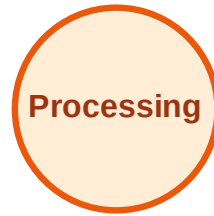
BFS Traversal

Step 20: Process node E

Legend



Complete



Processing



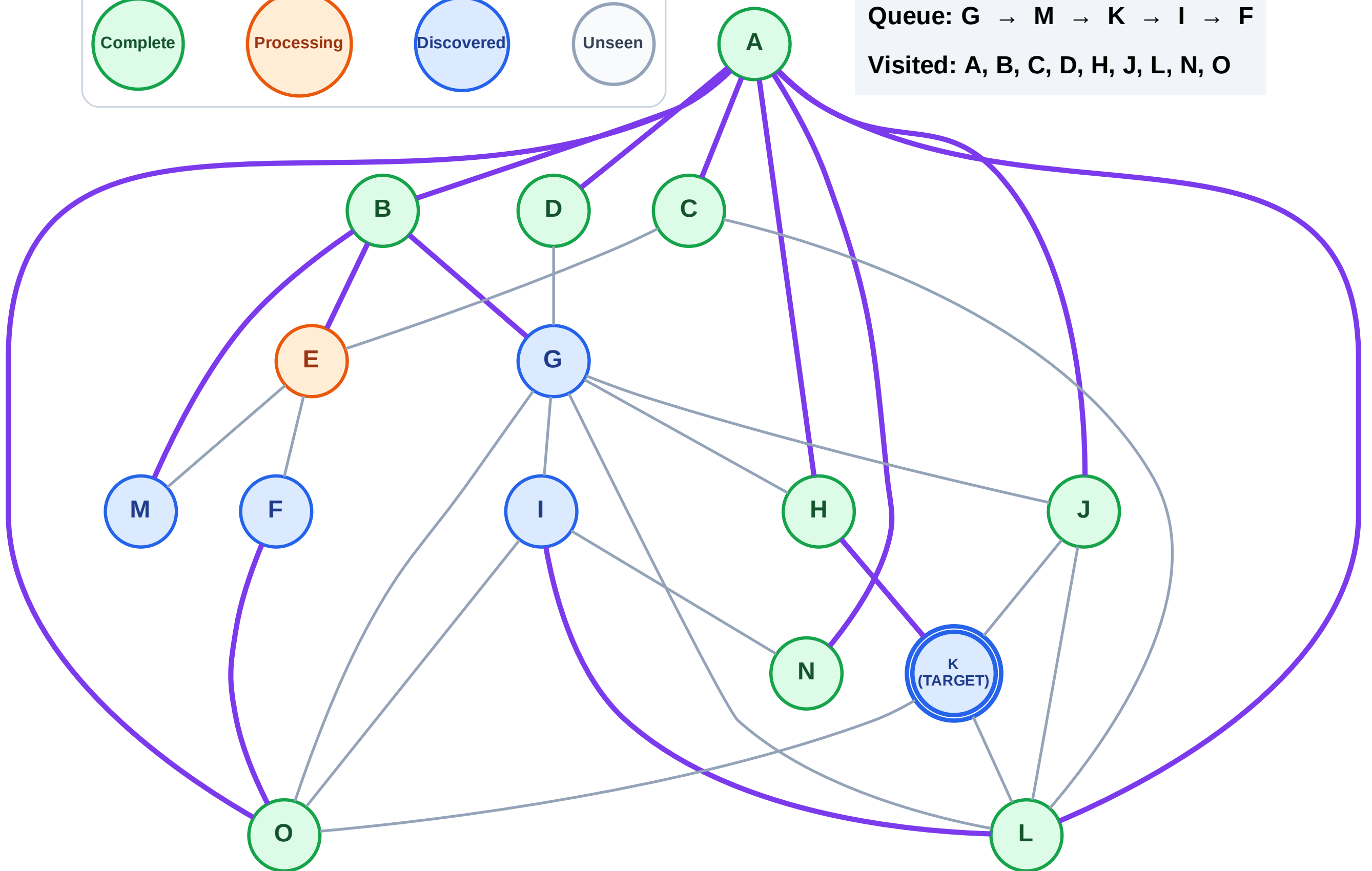
Discovered



Unseen

Queue: G → M → K → I → F

Visited: A, B, C, D, H, J, L, N, O



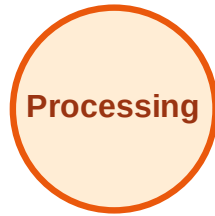
BFS Traversal

Step 21: No new neighbors from E

Legend



Complete



Processing



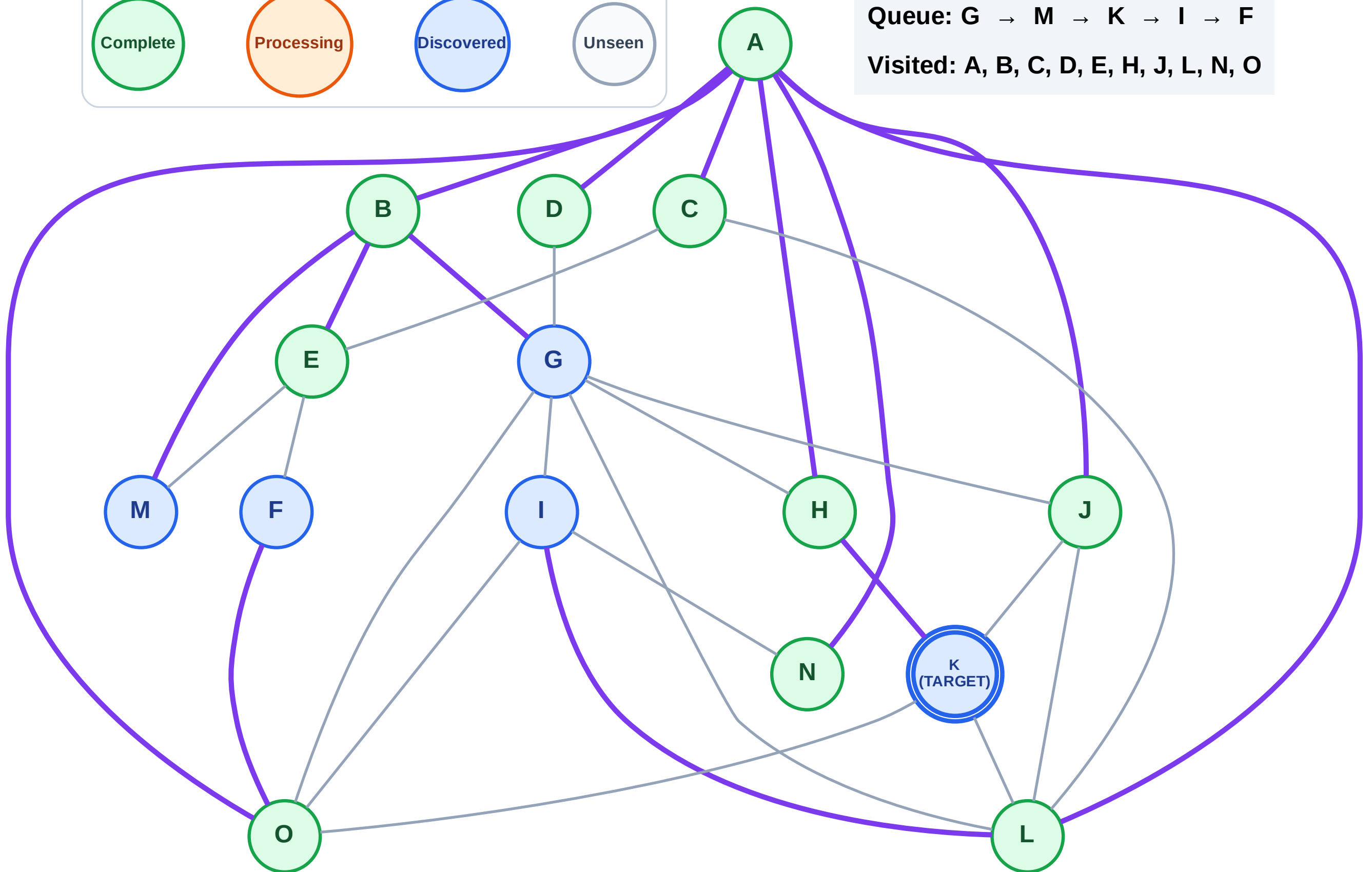
Discovered



Unseen

Queue: G → M → K → I → F

Visited: A, B, C, D, E, H, J, L, N, O



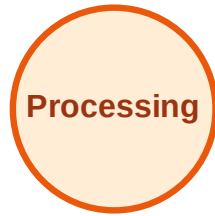
BFS Traversal

Step 22: Process node G

Legend



Complete



Processing



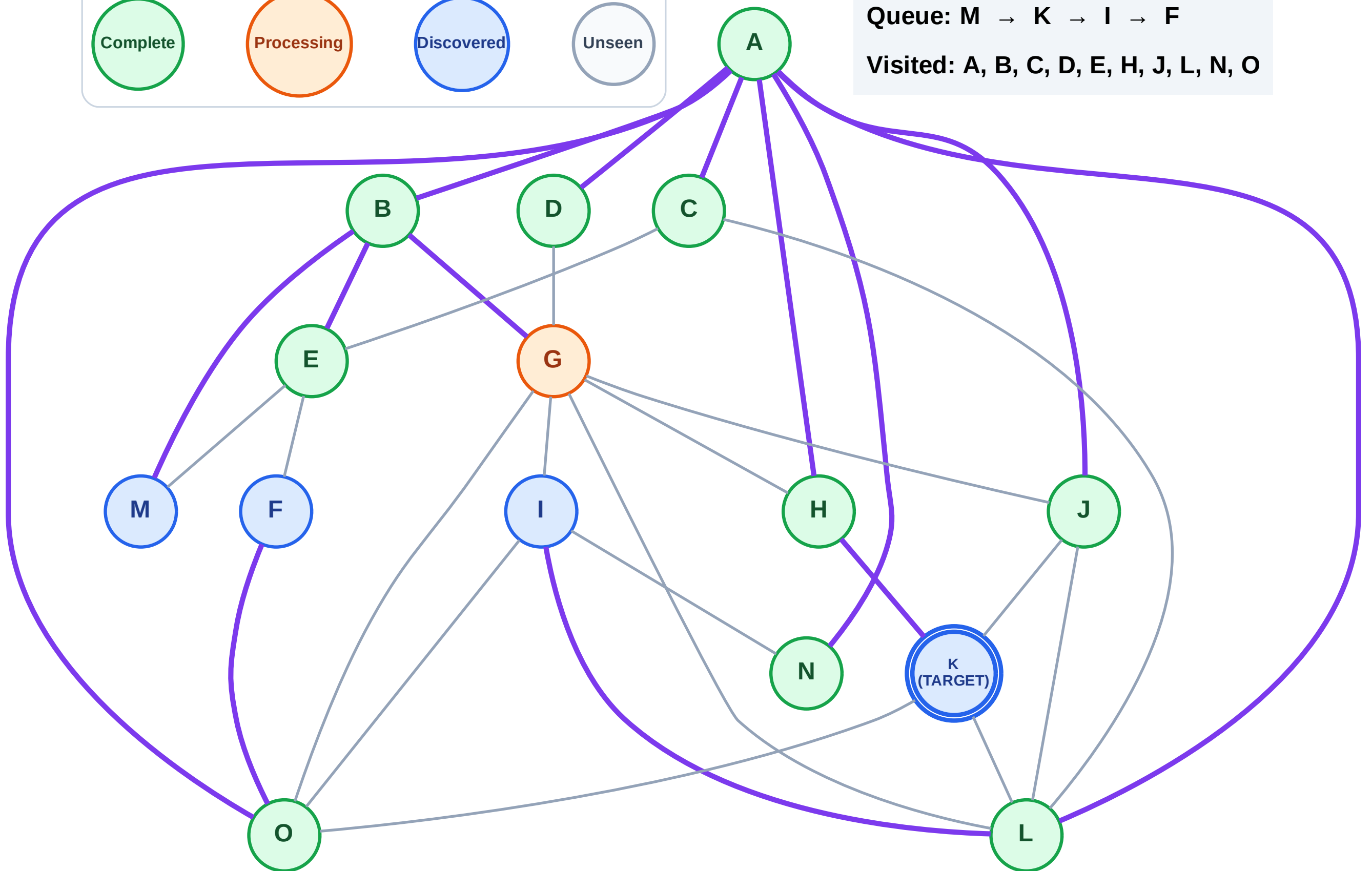
Discovered



Unseen

Queue: M → K → I → F

Visited: A, B, C, D, E, H, J, L, N, O



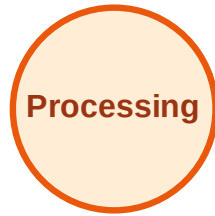
BFS Traversal

Step 23: No new neighbors from G

Legend



Complete



Processing



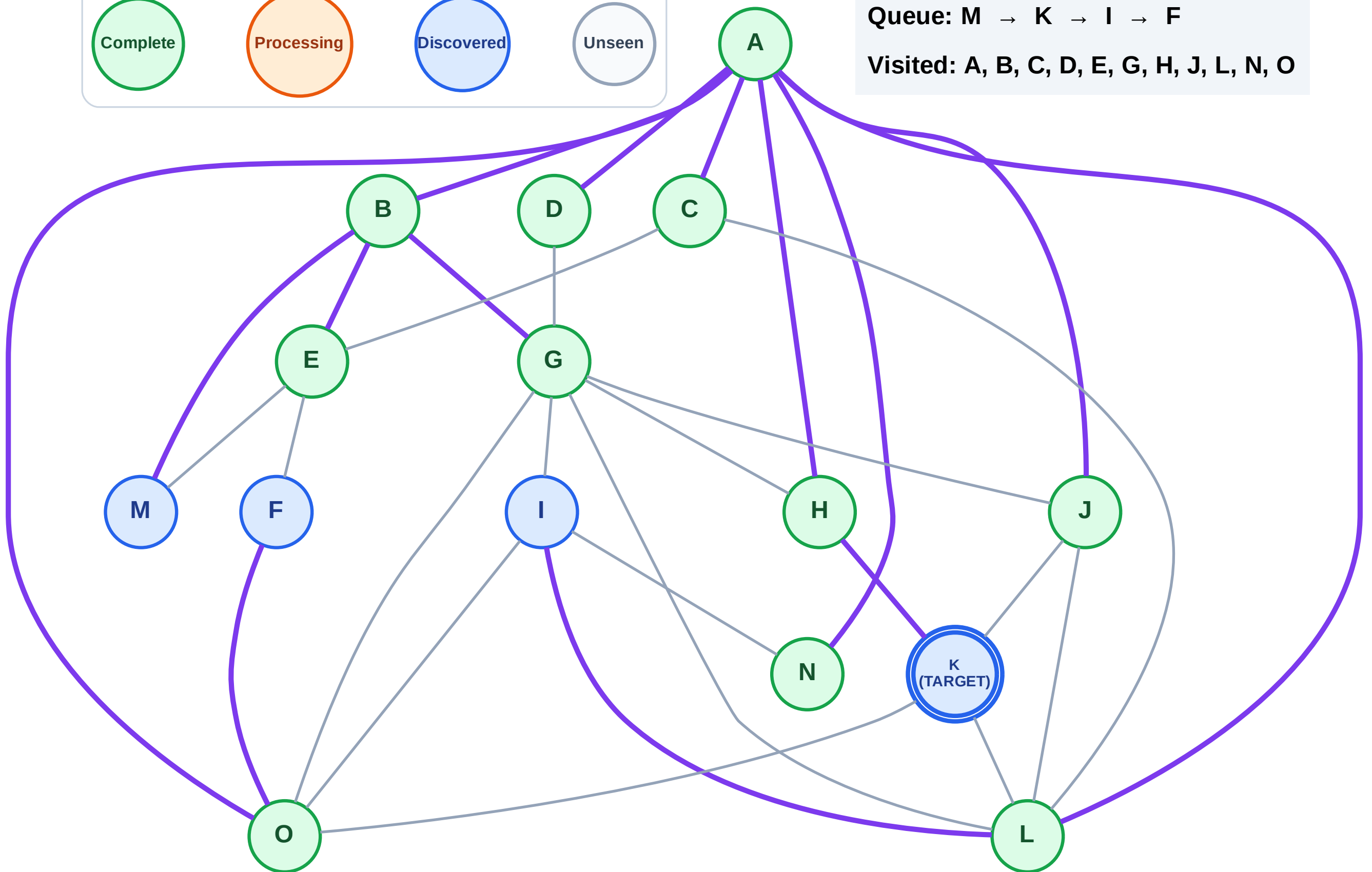
Discovered



Unseen

Queue: M → K → I → F

Visited: A, B, C, D, E, G, H, J, L, N, O



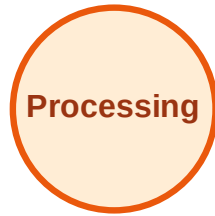
BFS Traversal

Step 24: Process node M

Legend



Complete



Processing



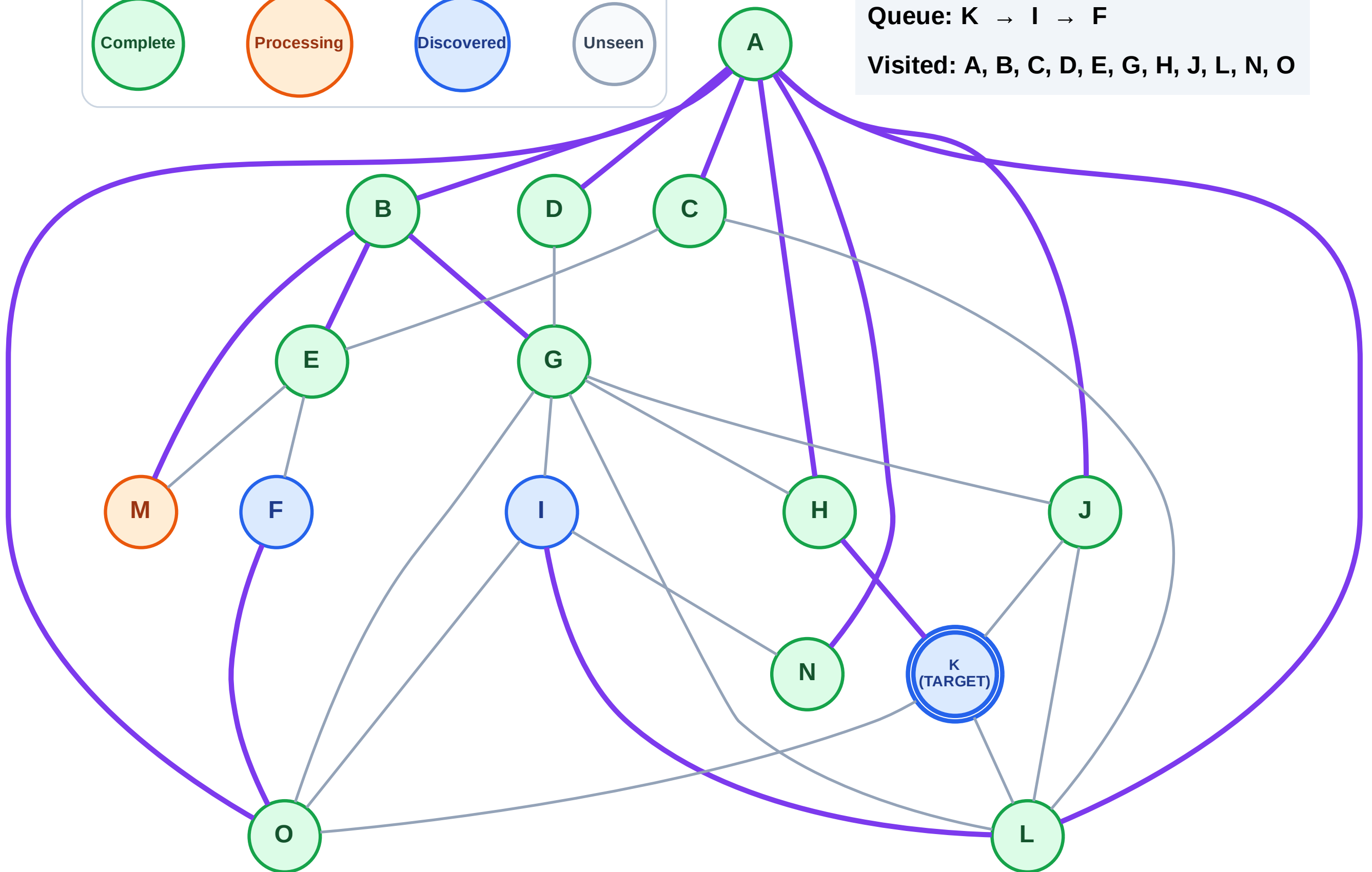
Discovered



Unseen

Queue: K → I → F

Visited: A, B, C, D, E, G, H, J, L, N, O



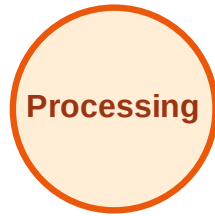
BFS Traversal

Step 25: No new neighbors from M

Legend



Complete



Processing



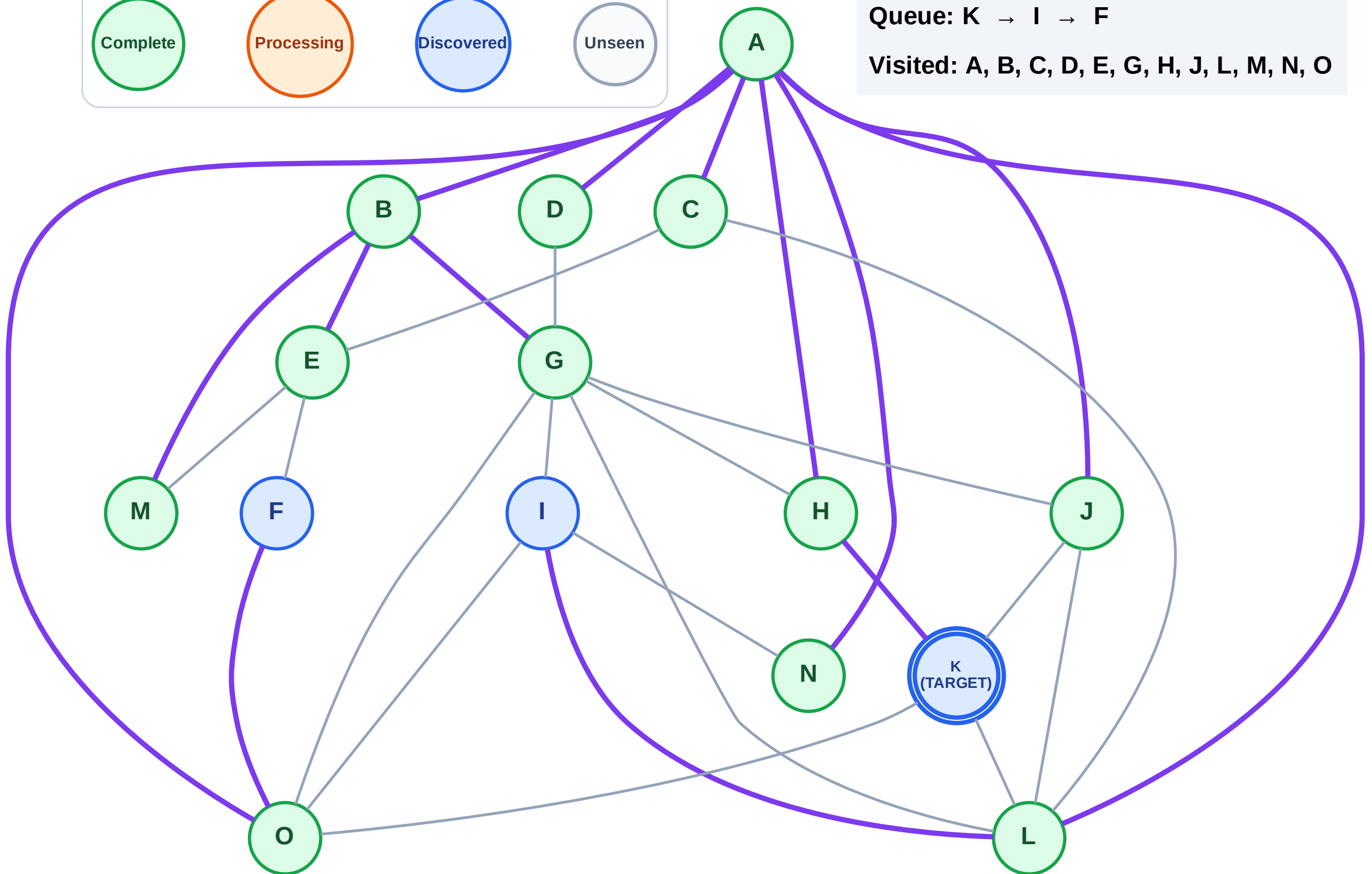
Discovered



Unseen

Queue: K → I → F

Visited: A, B, C, D, E, G, H, J, L, M, N, O



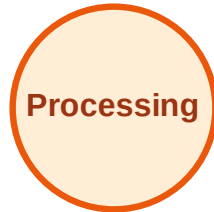
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



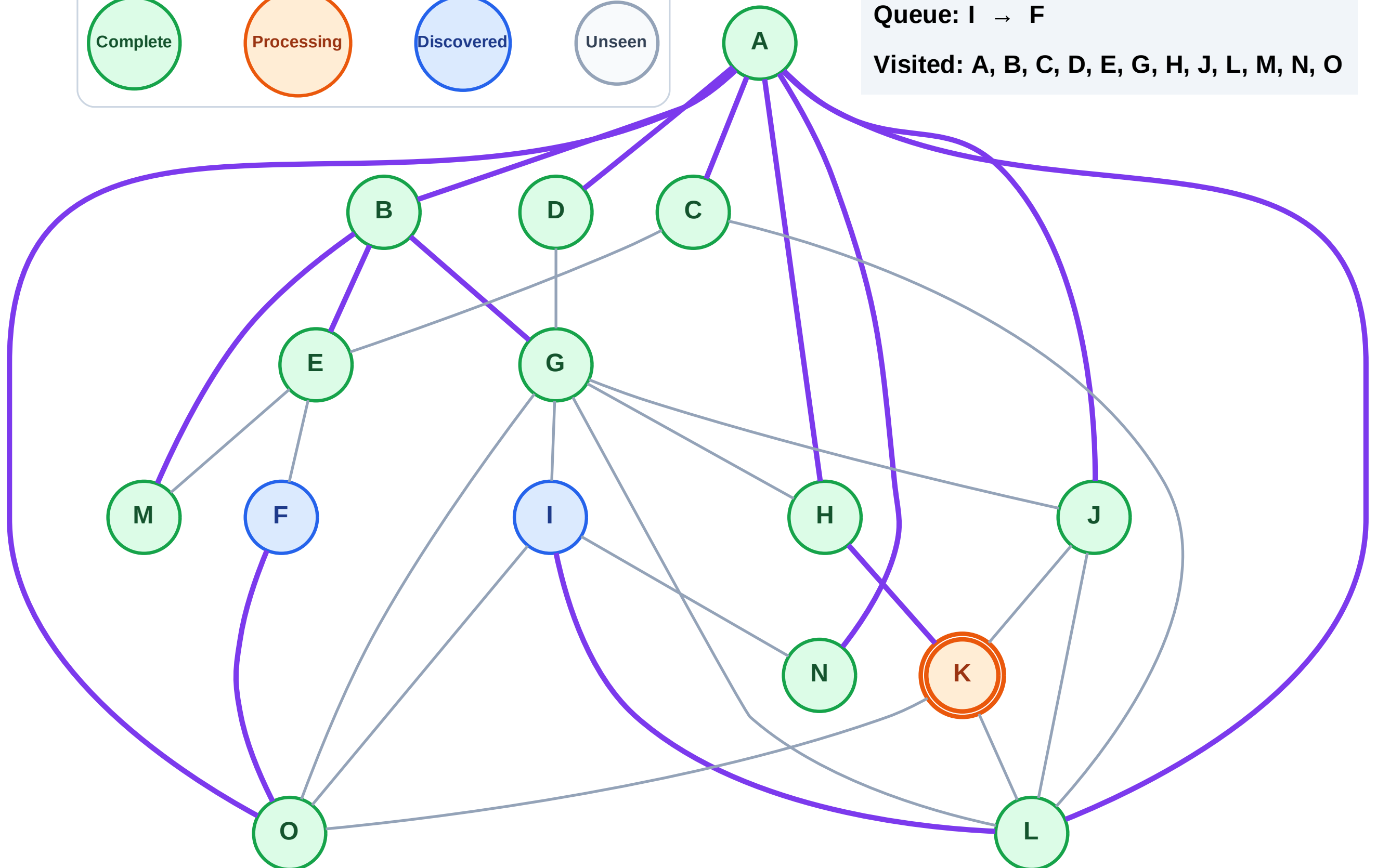
Discovered



Unseen

Queue: I → F

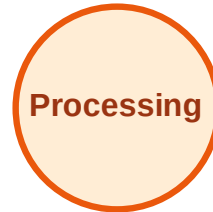
Visited: A, B, C, D, E, G, H, J, L, M, N, O



BFS Traversal

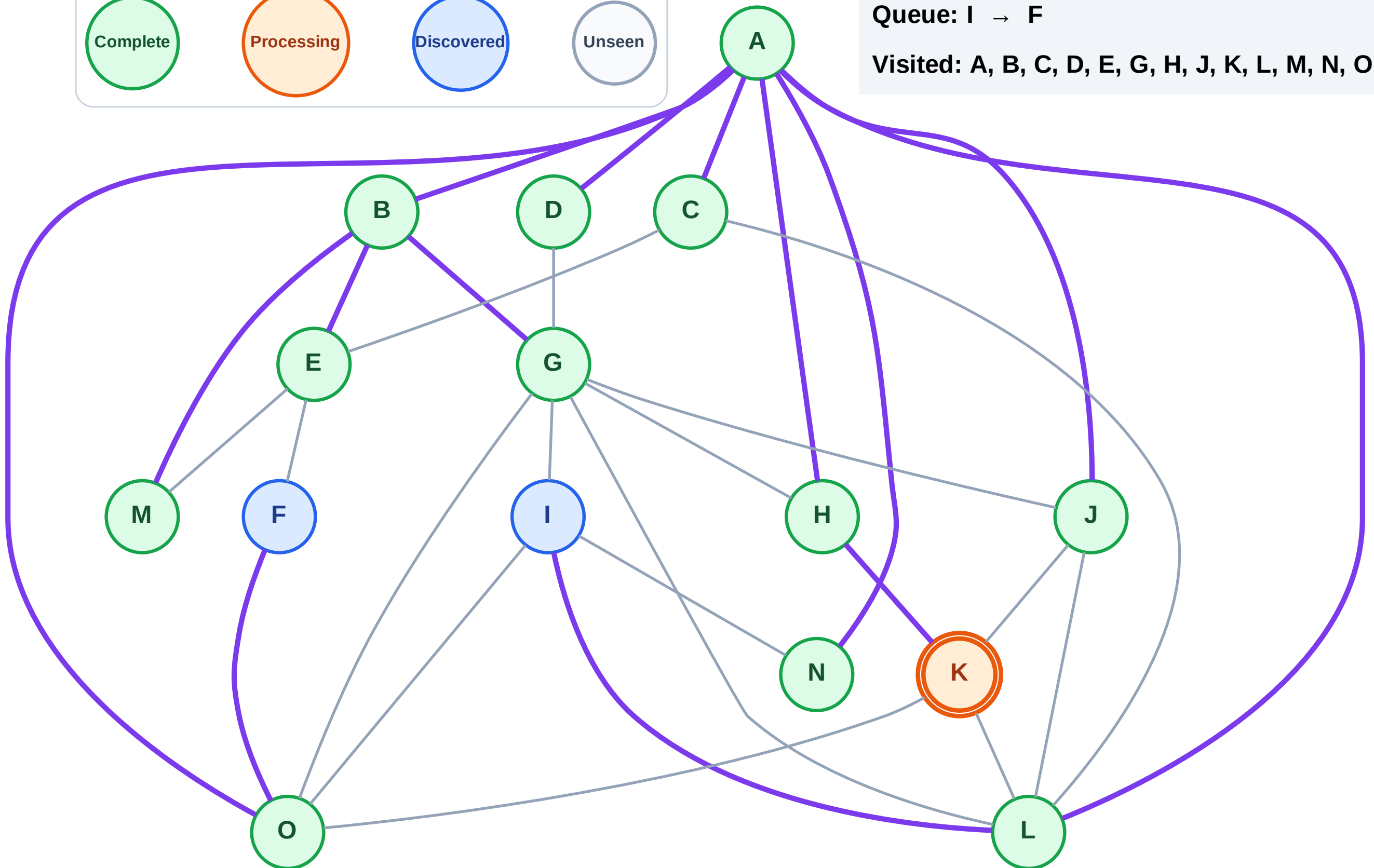
Step 27: TARGET NODE K REACHED! Path (2 hops): A → H → K

Legend



Queue: I → F

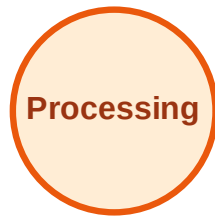
Visited: A, B, C, D, E, G, H, J, K, L, M, N, O



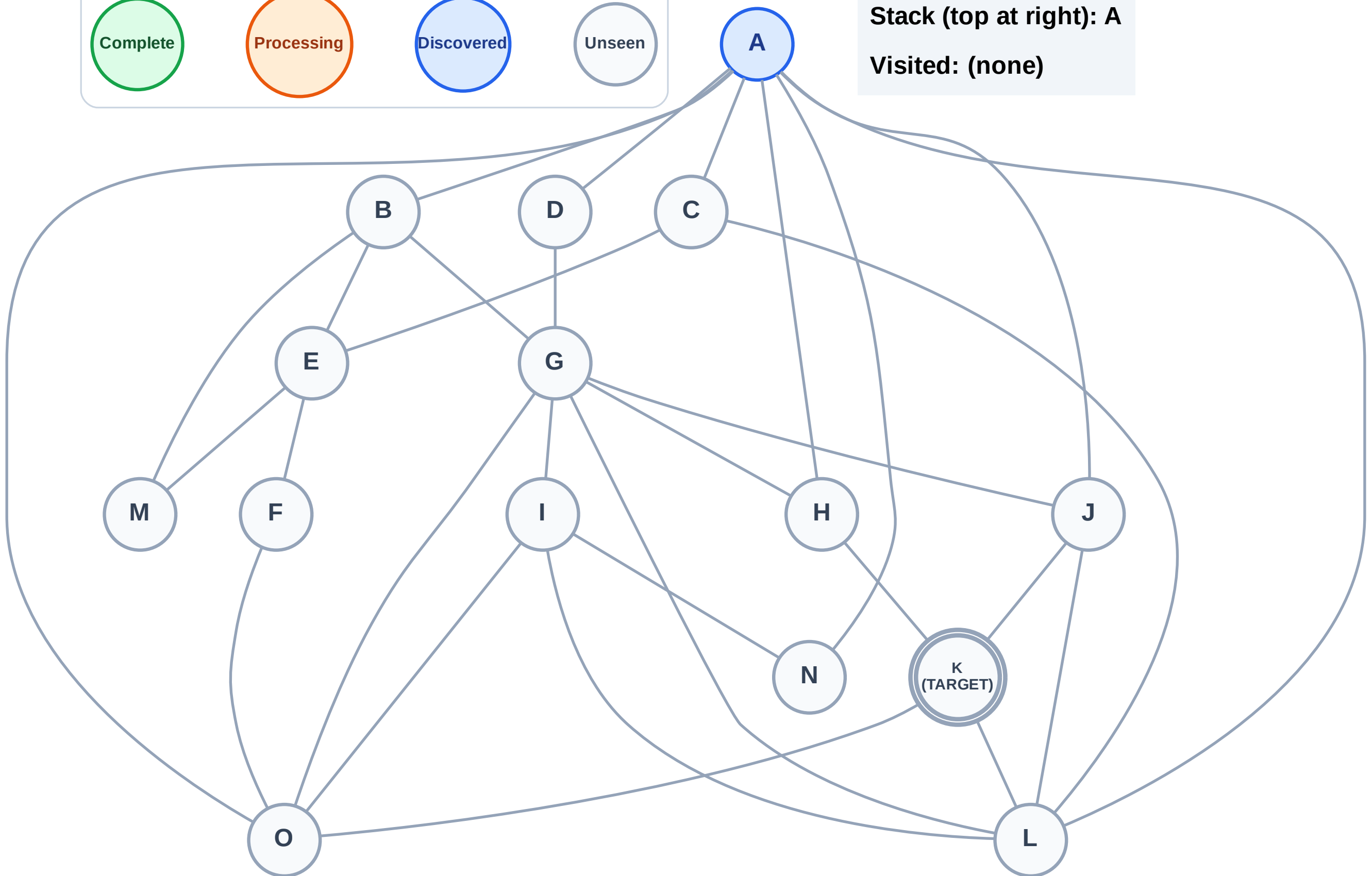
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



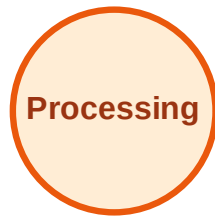
Stack (top at right): A
Visited: (none)



DFS Traversal

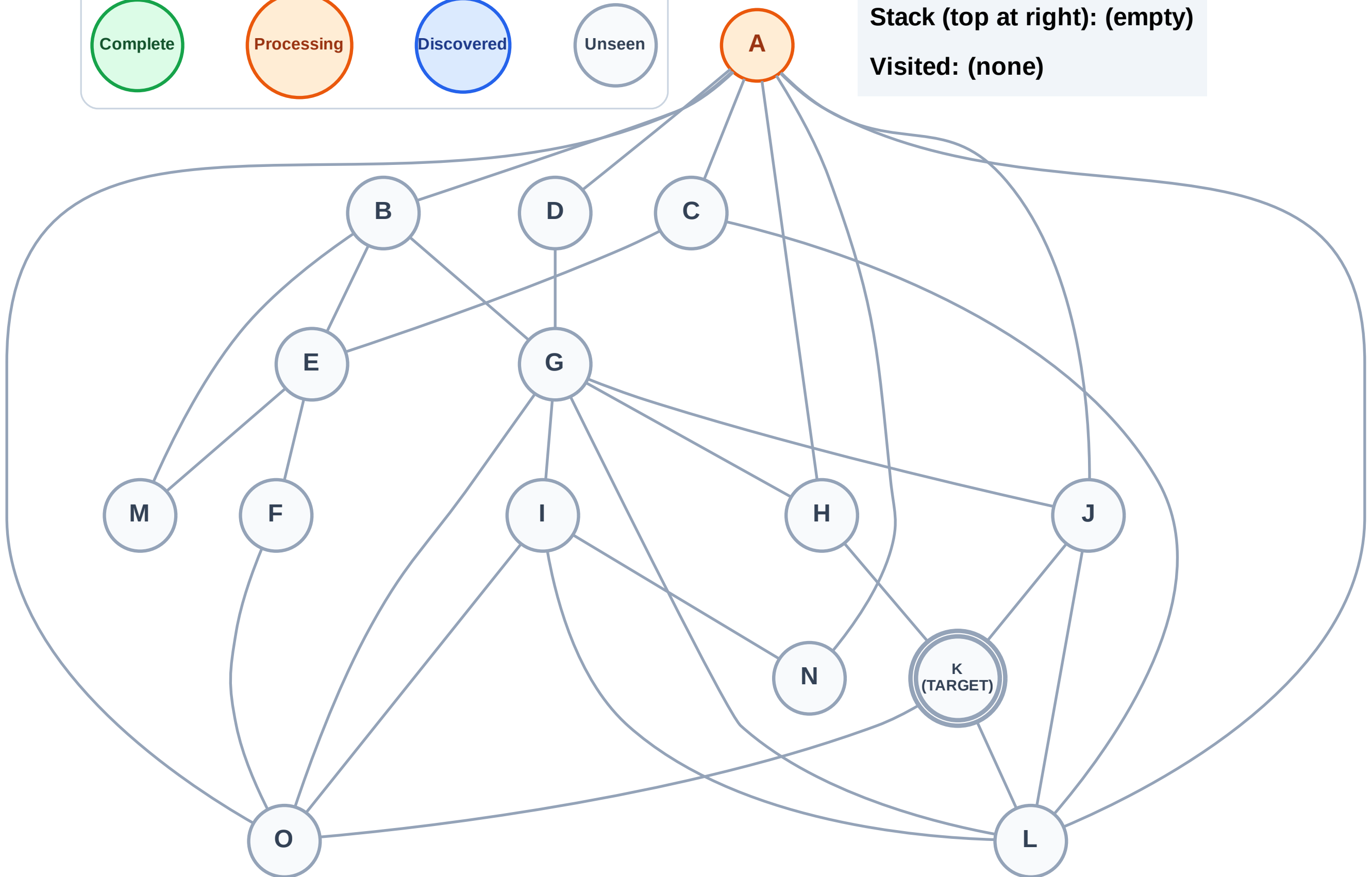
Step 2: Process node A

Legend



Stack (top at right): (empty)

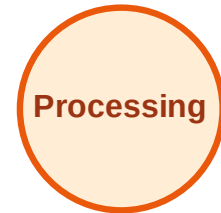
Visited: (none)



DFS Traversal

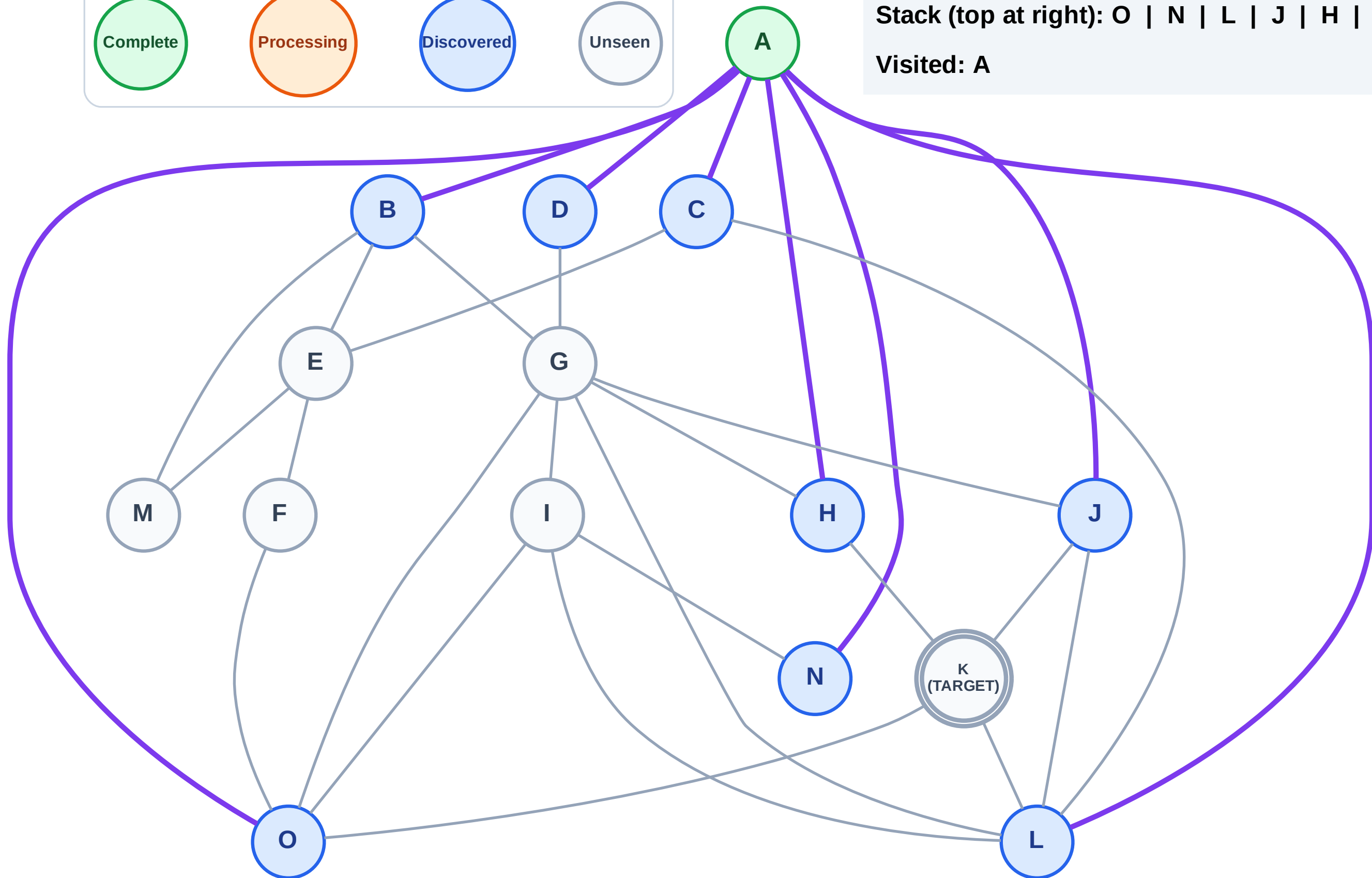
Step 3: Discover O, N, L, J, H, D, C, B from A

Legend



Stack (top at right): O | N | L | J | H | D | C | B

Visited: A



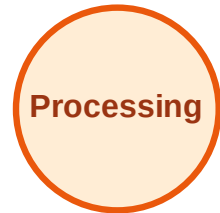
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



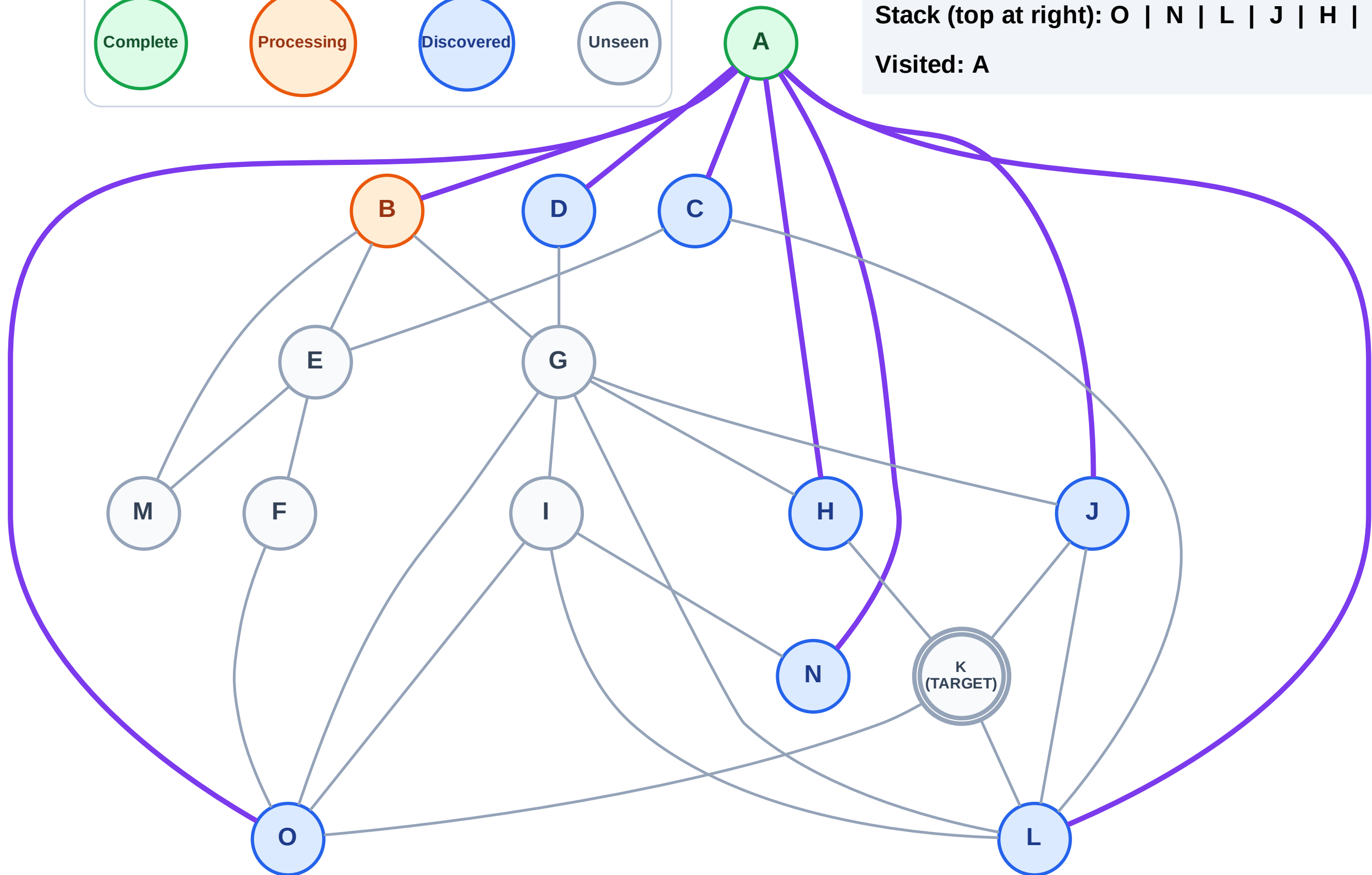
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C

Visited: A



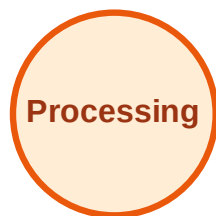
DFS Traversal

Step 5: Discover M, G, E from B

Legend



Complete



Processing



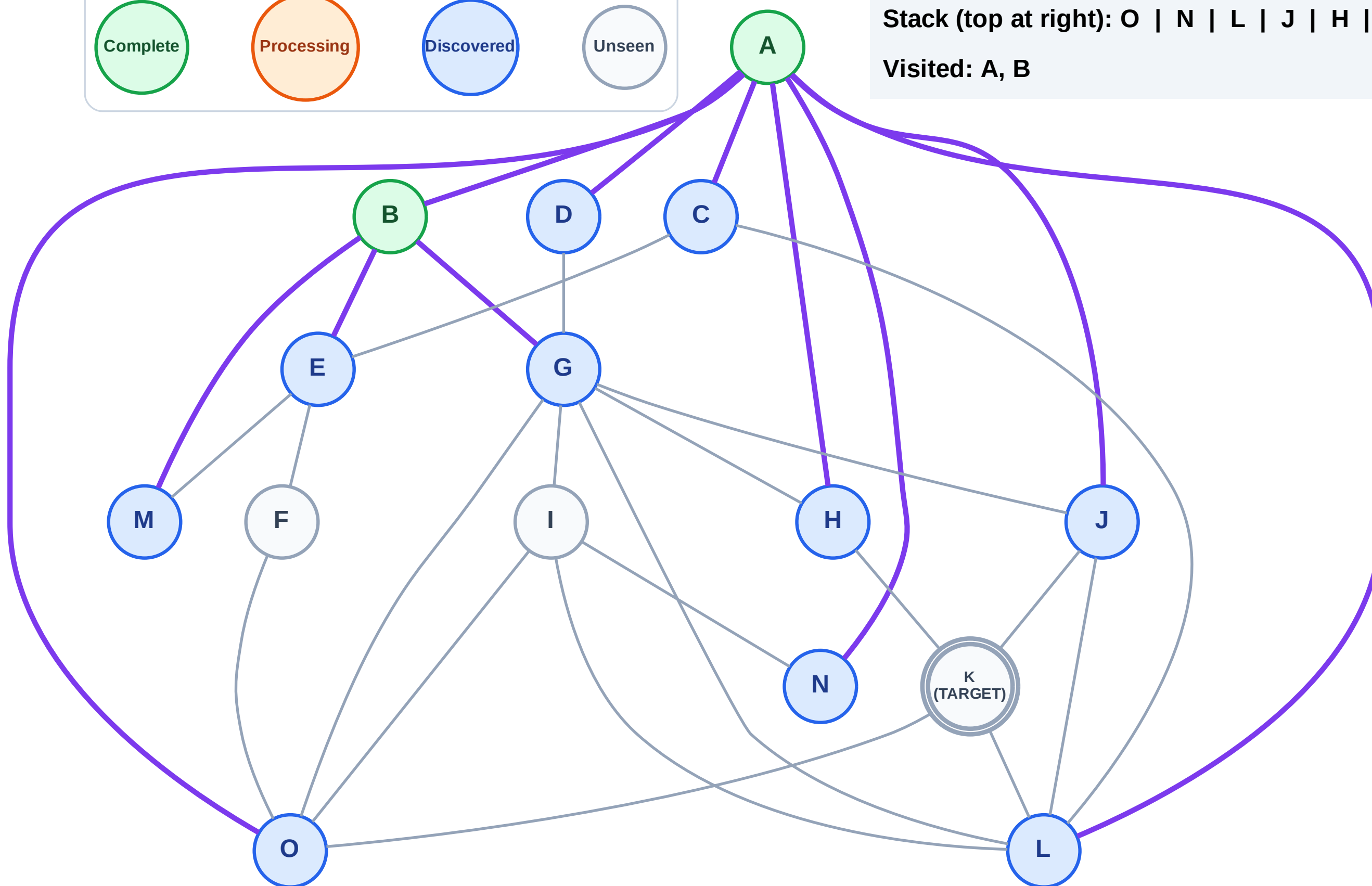
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

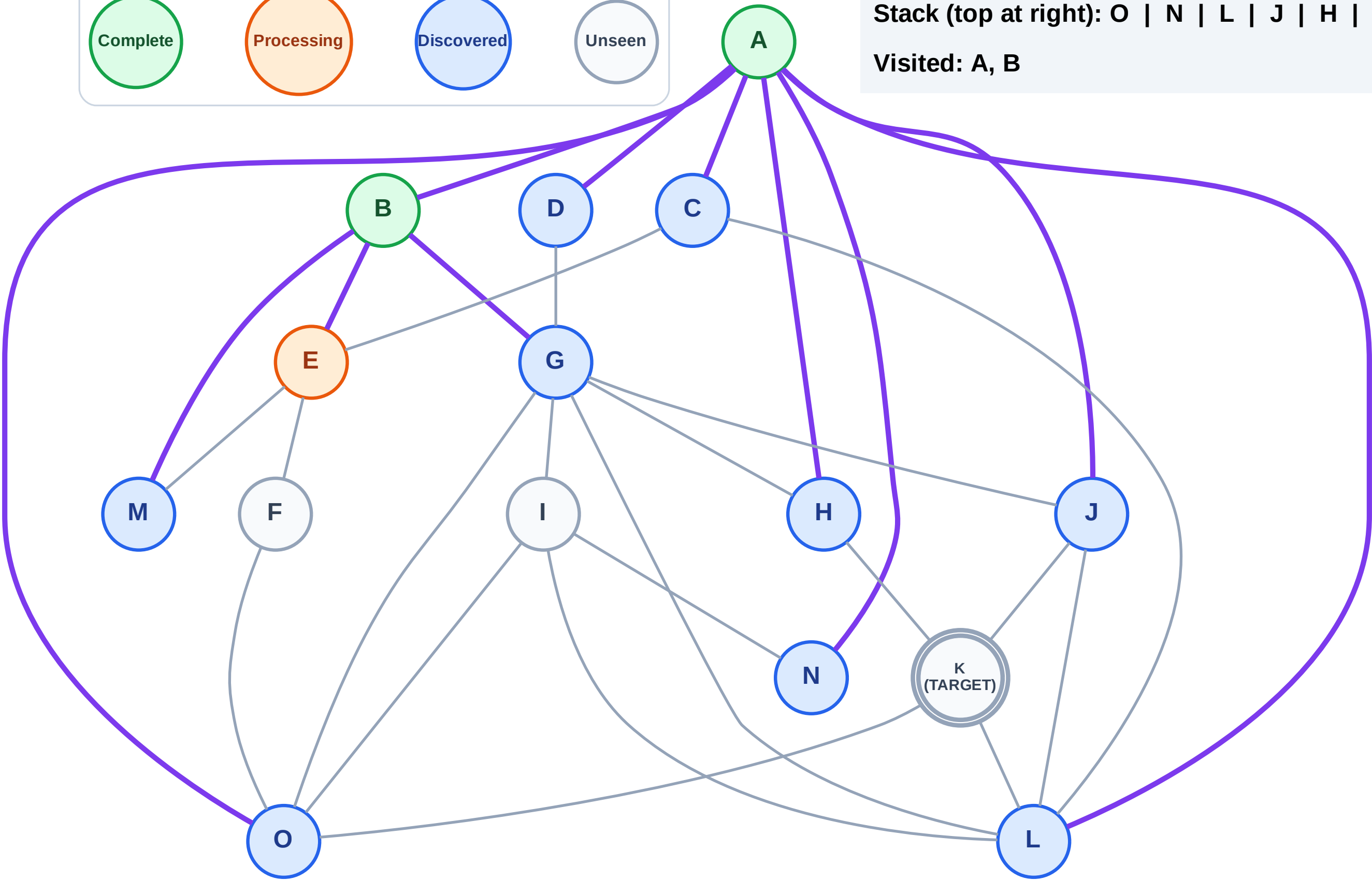
Processing

Discovered

Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G

Visited: A, B



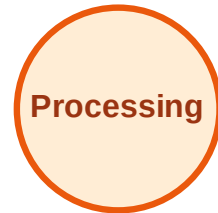
DFS Traversal

Step 7: Discover F from E

Legend



Complete



Processing



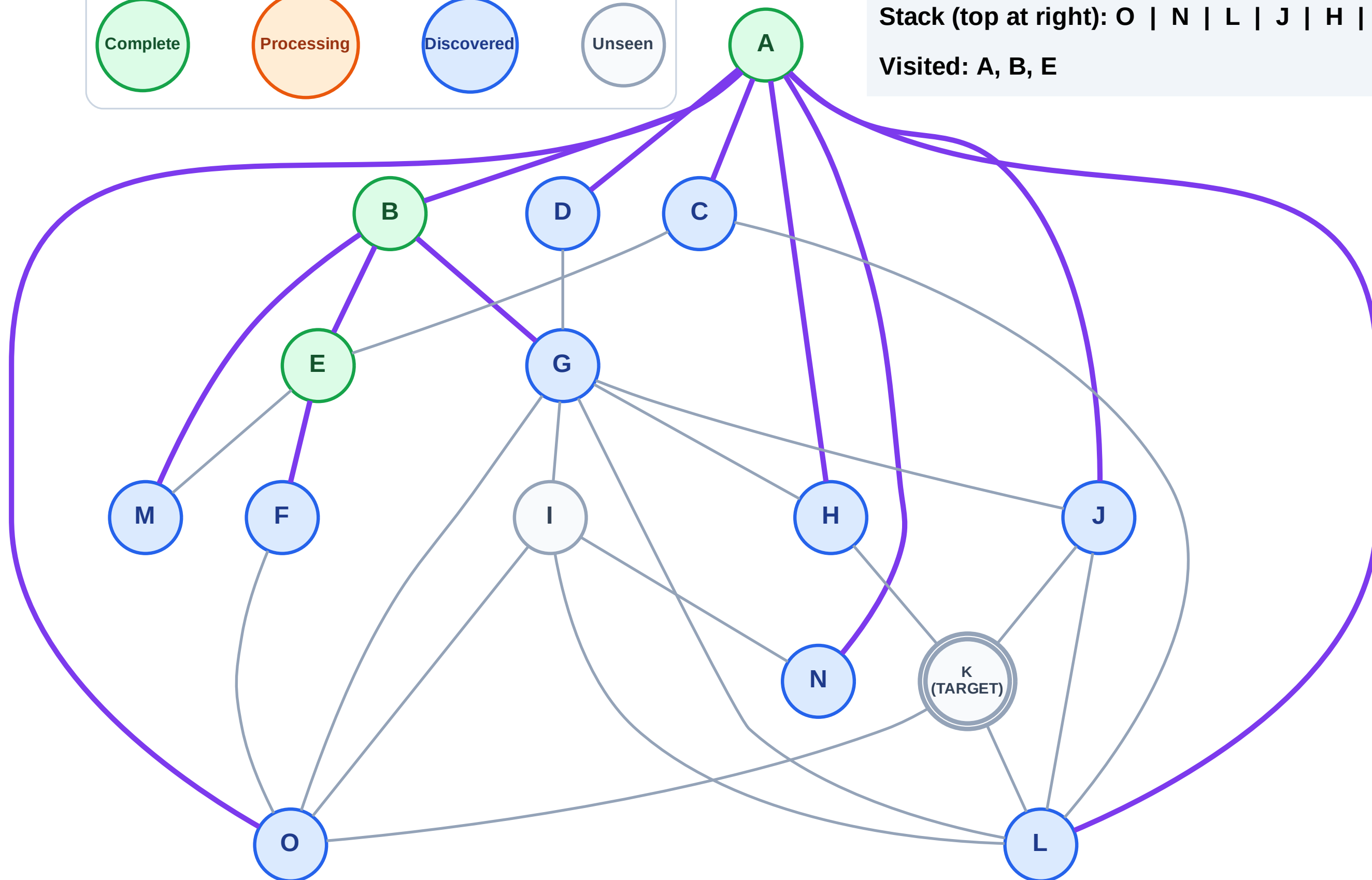
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

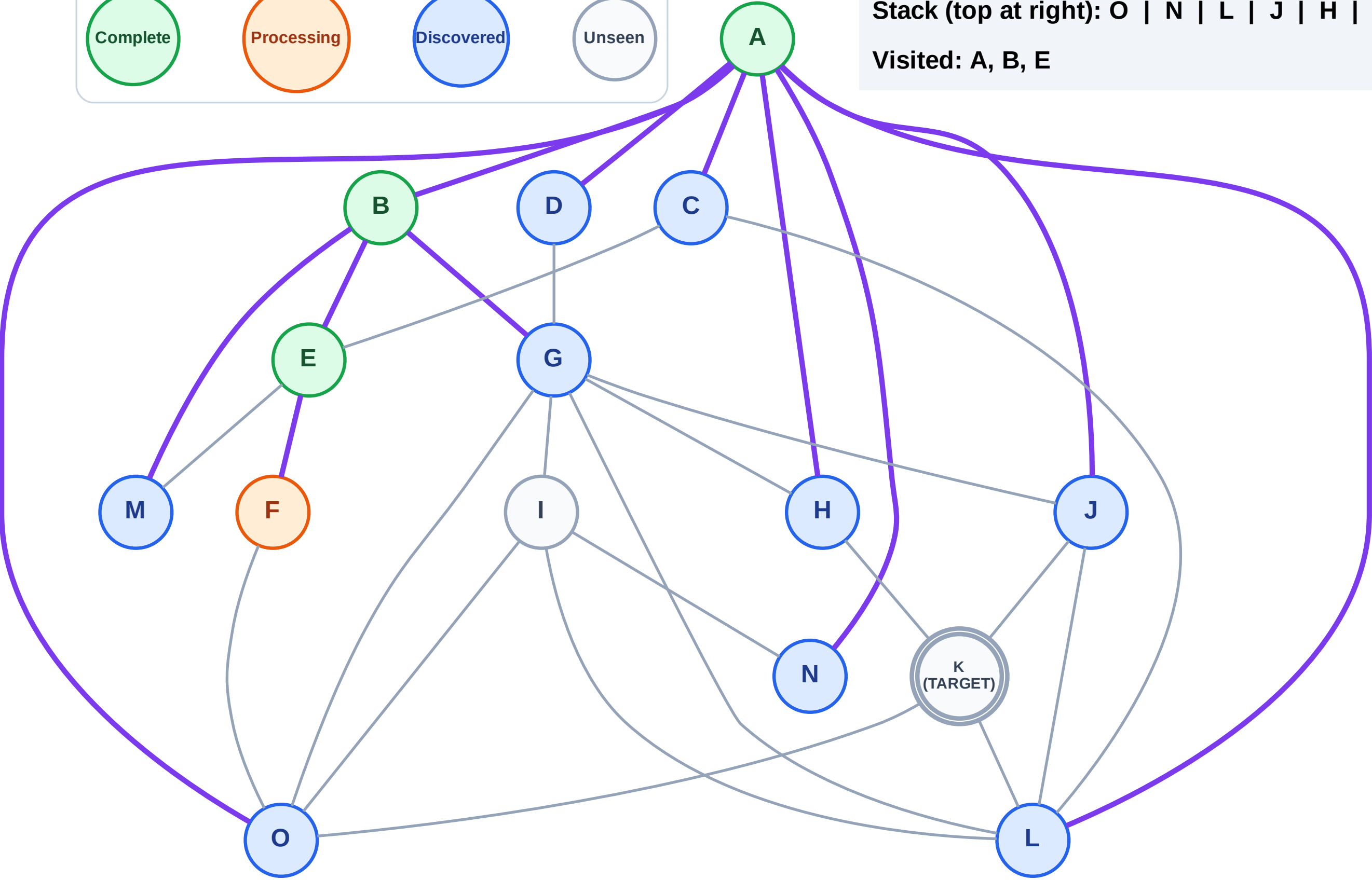
Processing

Discovered

Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G

Visited: A, B, E

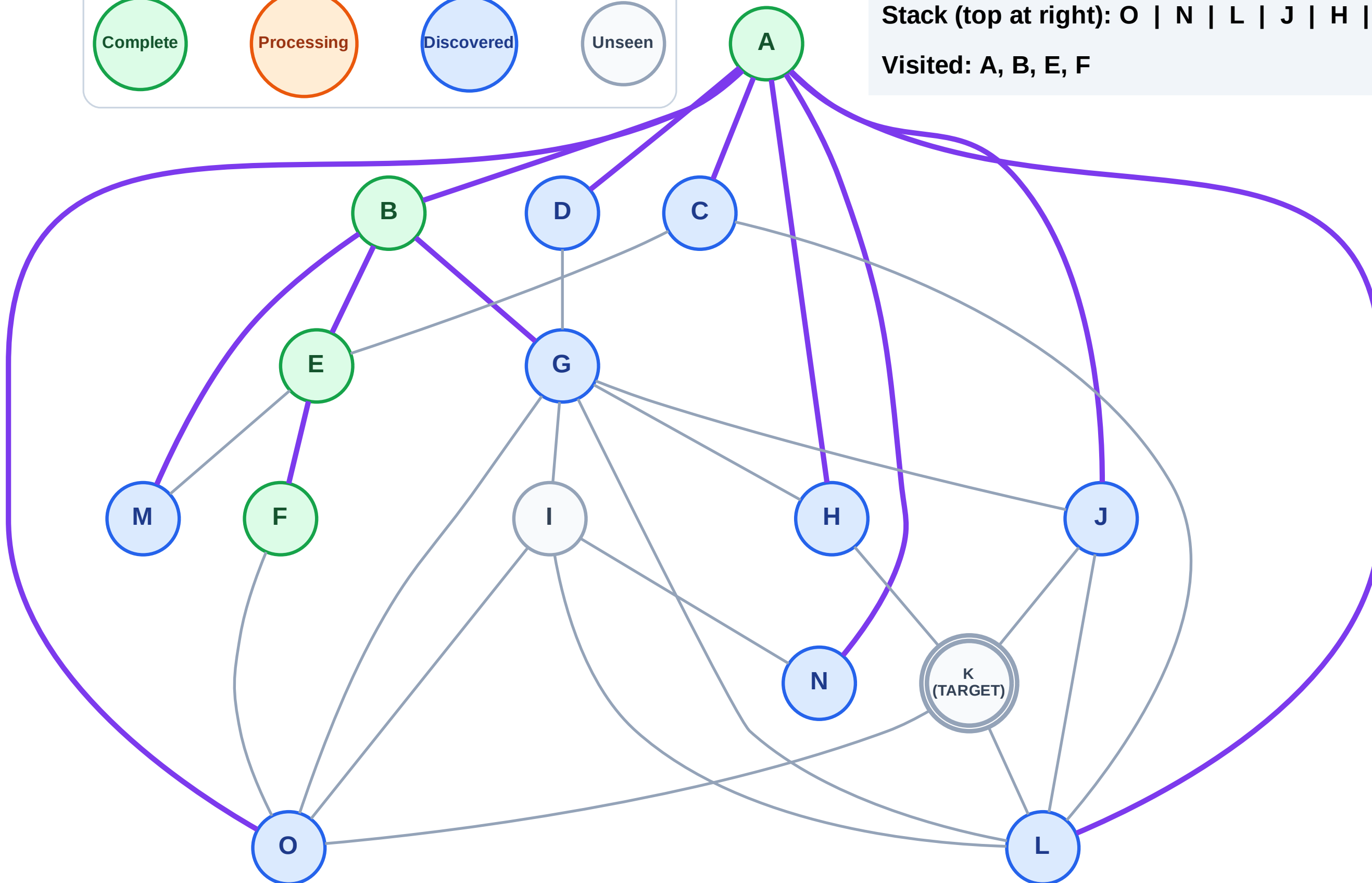


Step 9: No new neighbors from F

Legend

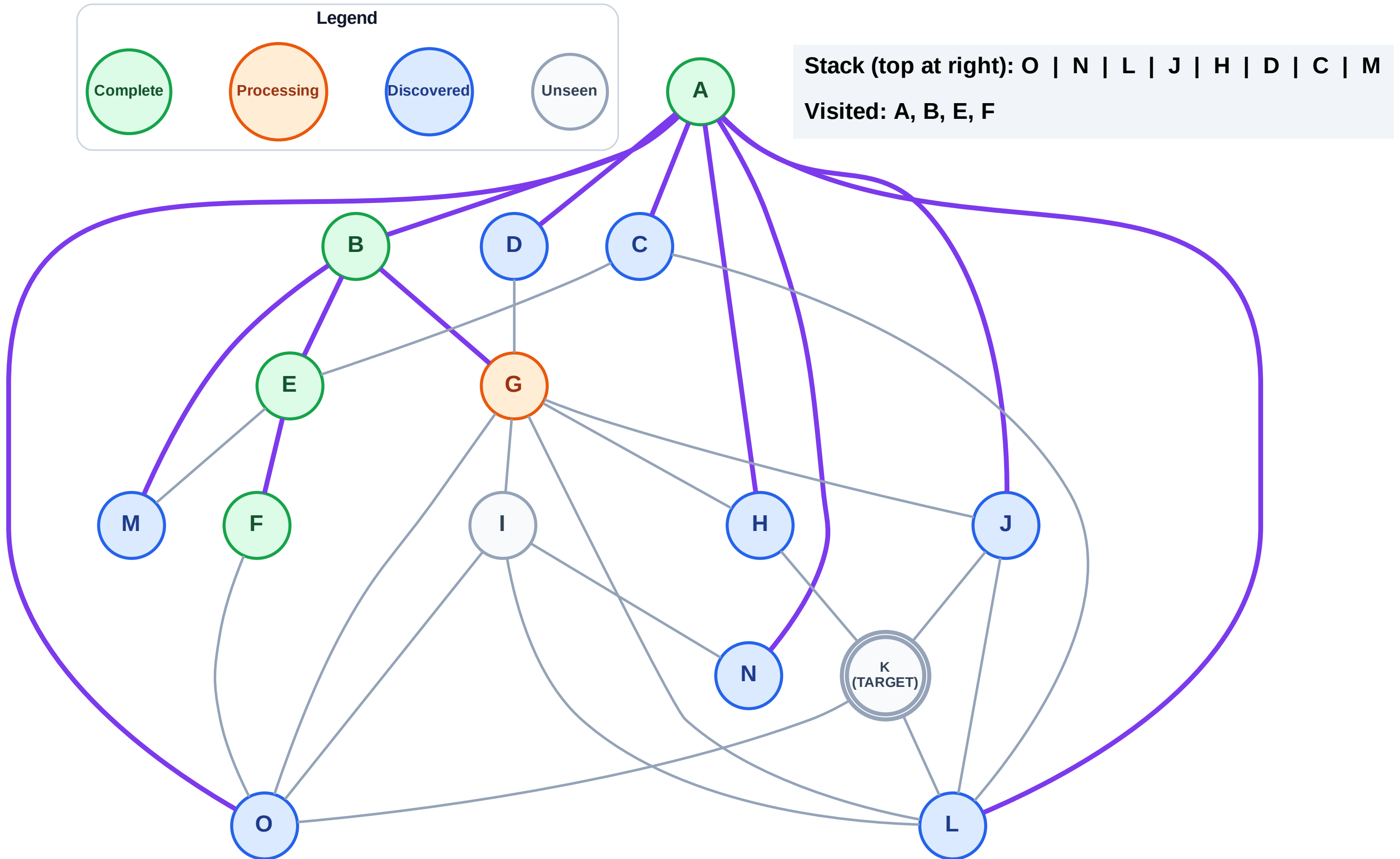
Unseen

Visited: A, B, E, F



Step 10: Process node G

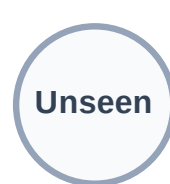
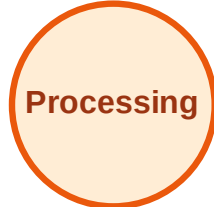
Step 10: Process node G



DFS Traversal

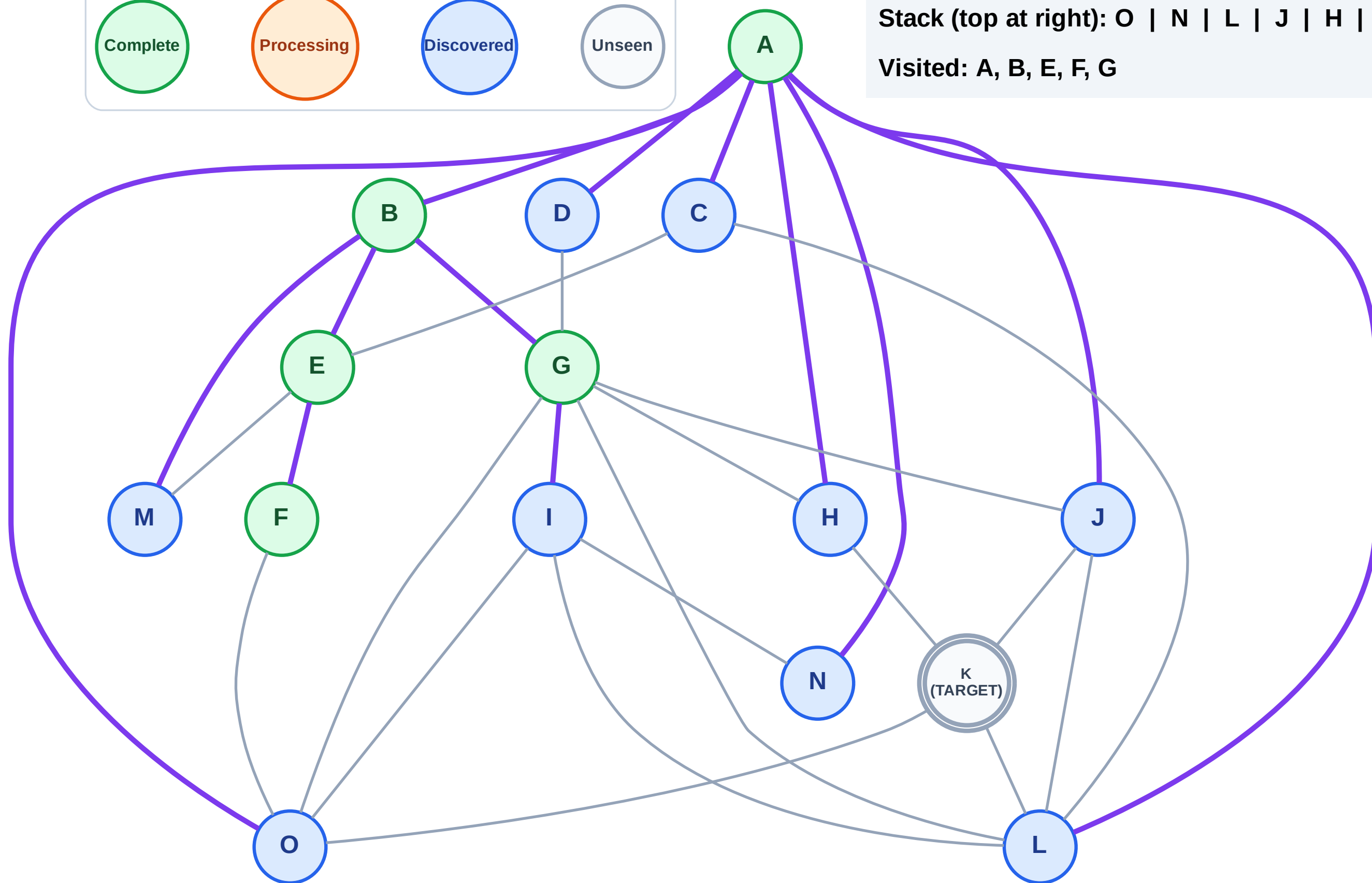
Step 11: Discover I from G

Legend



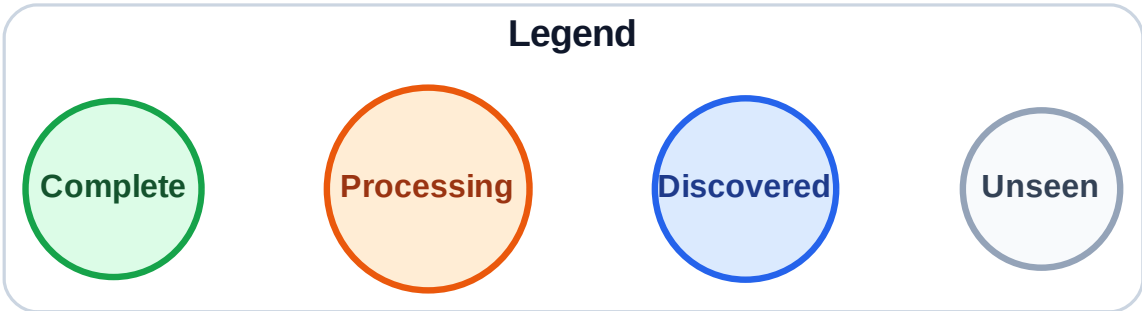
Stack (top at right): O | N | L | J | H | D | C | M | I

Visited: A, B, E, F, G



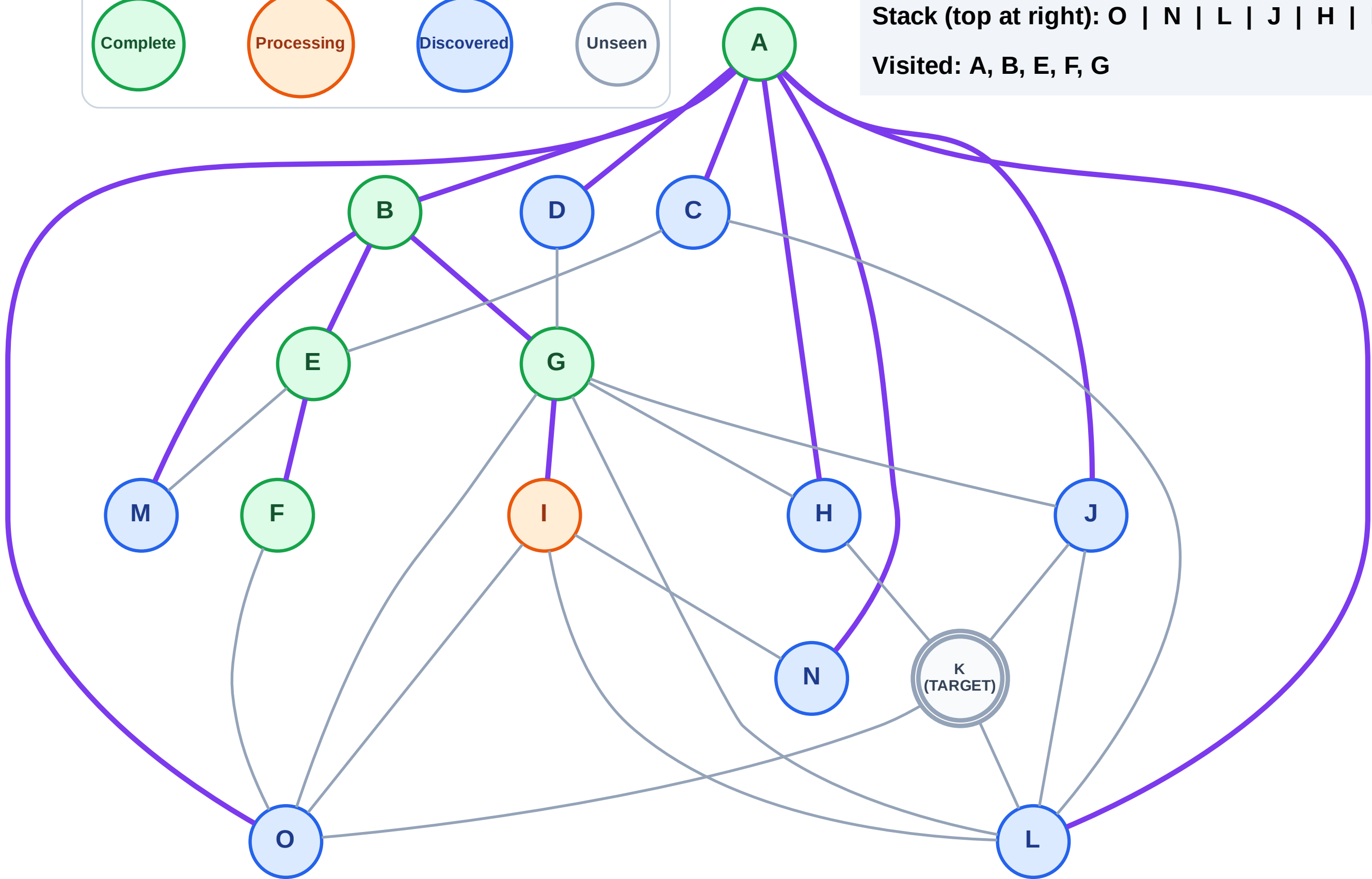
DFS Traversal

Step 12: Process node I



Stack (top at right): O | N | L | J | H | D | C | M

Visited: A, B, E, F, G



DFS Traversal

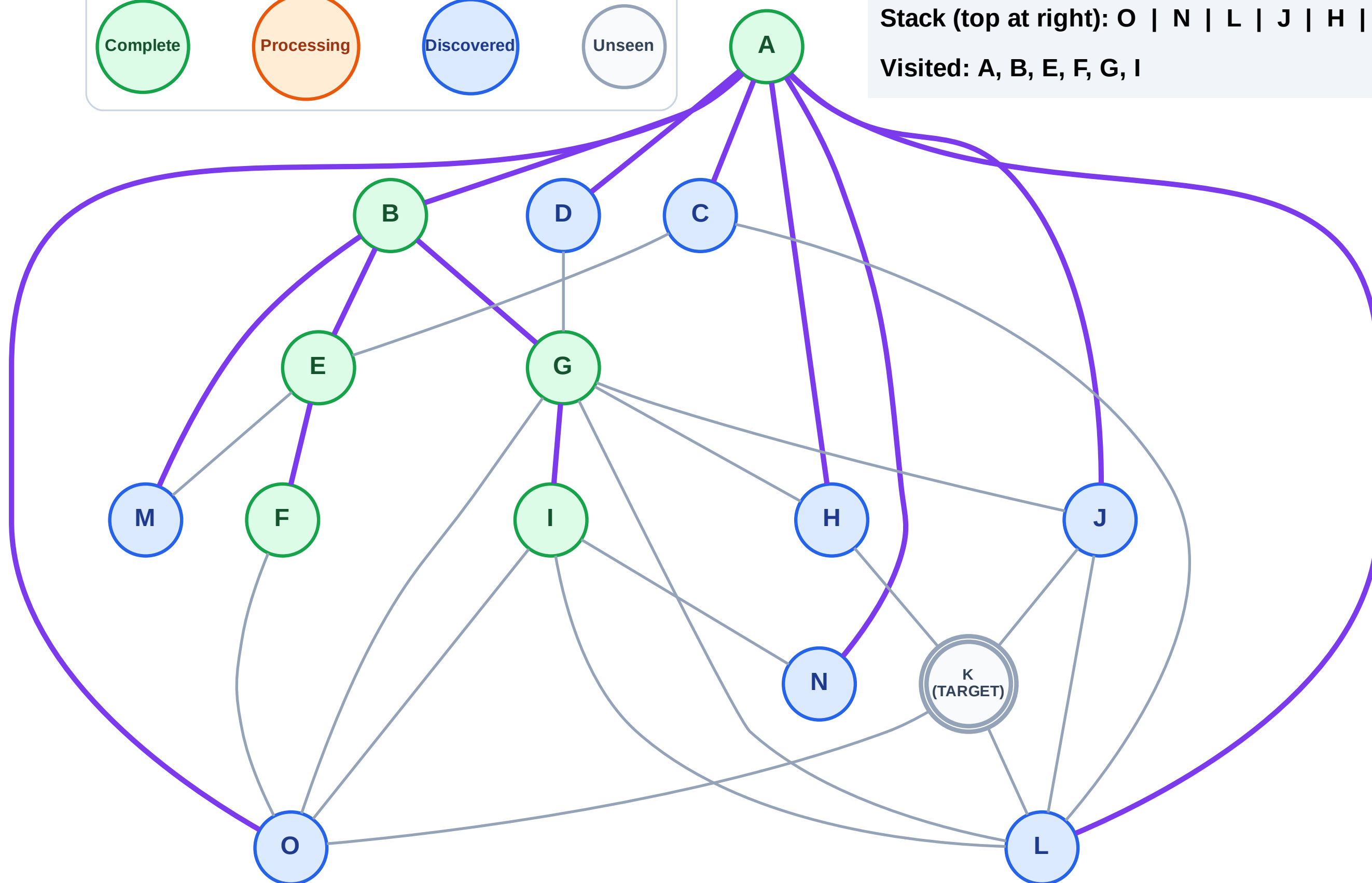
Step 13: No new neighbors from I

Legend



Stack (top at right): O | N | L | J | H | D | C | M

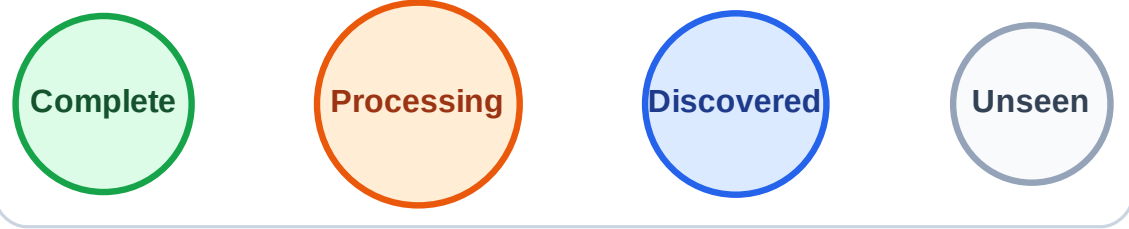
Visited: A, B, E, F, G, I



DFS Traversal

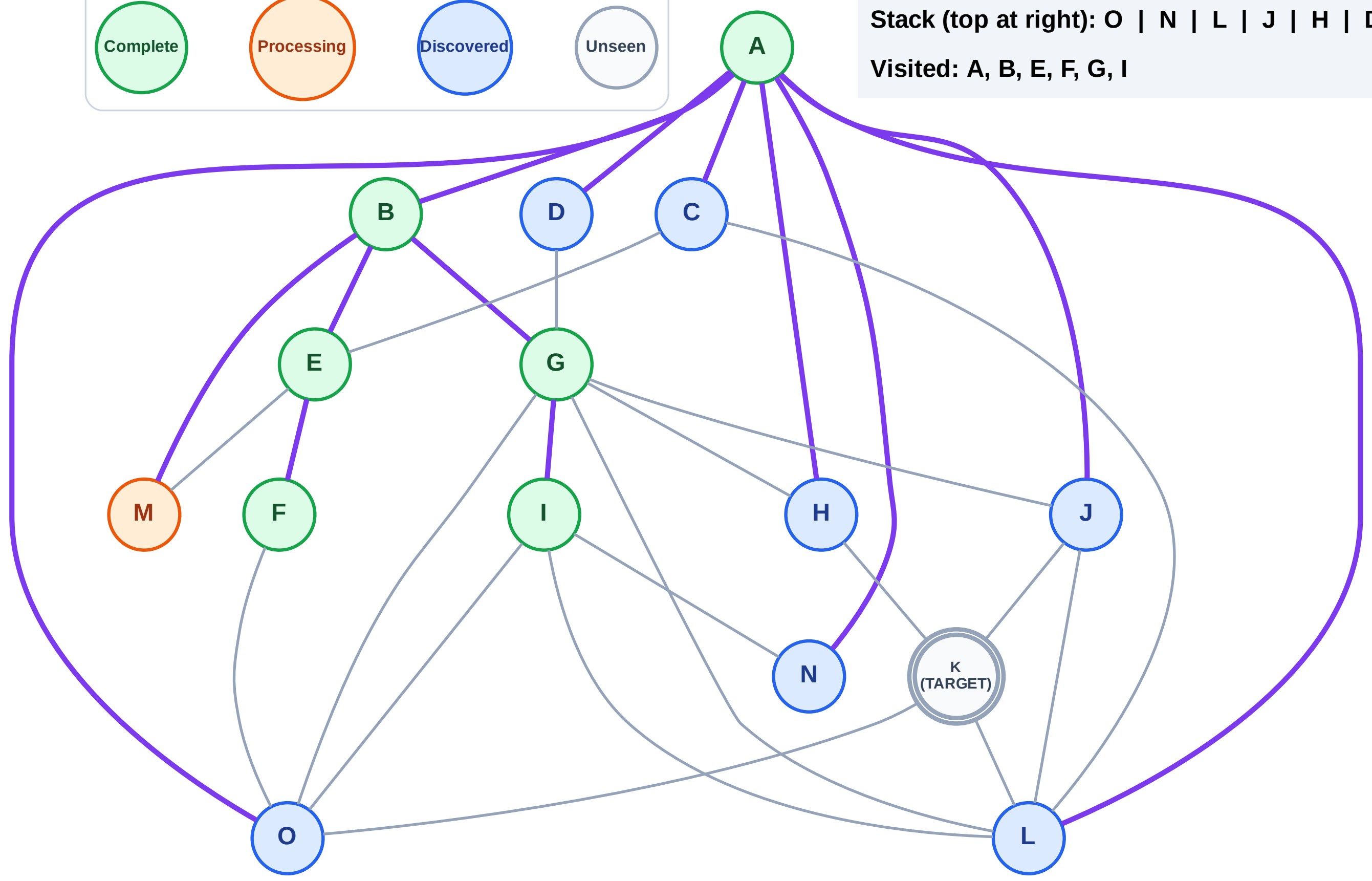
Step 14: Process node M

Legend



Stack (top at right): O | N | L | J | H | D | C

Visited: A, B, E, F, G, I



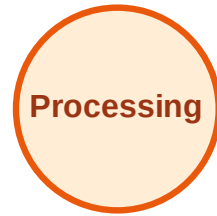
DFS Traversal

Step 15: No new neighbors from M

Legend



Complete



Processing



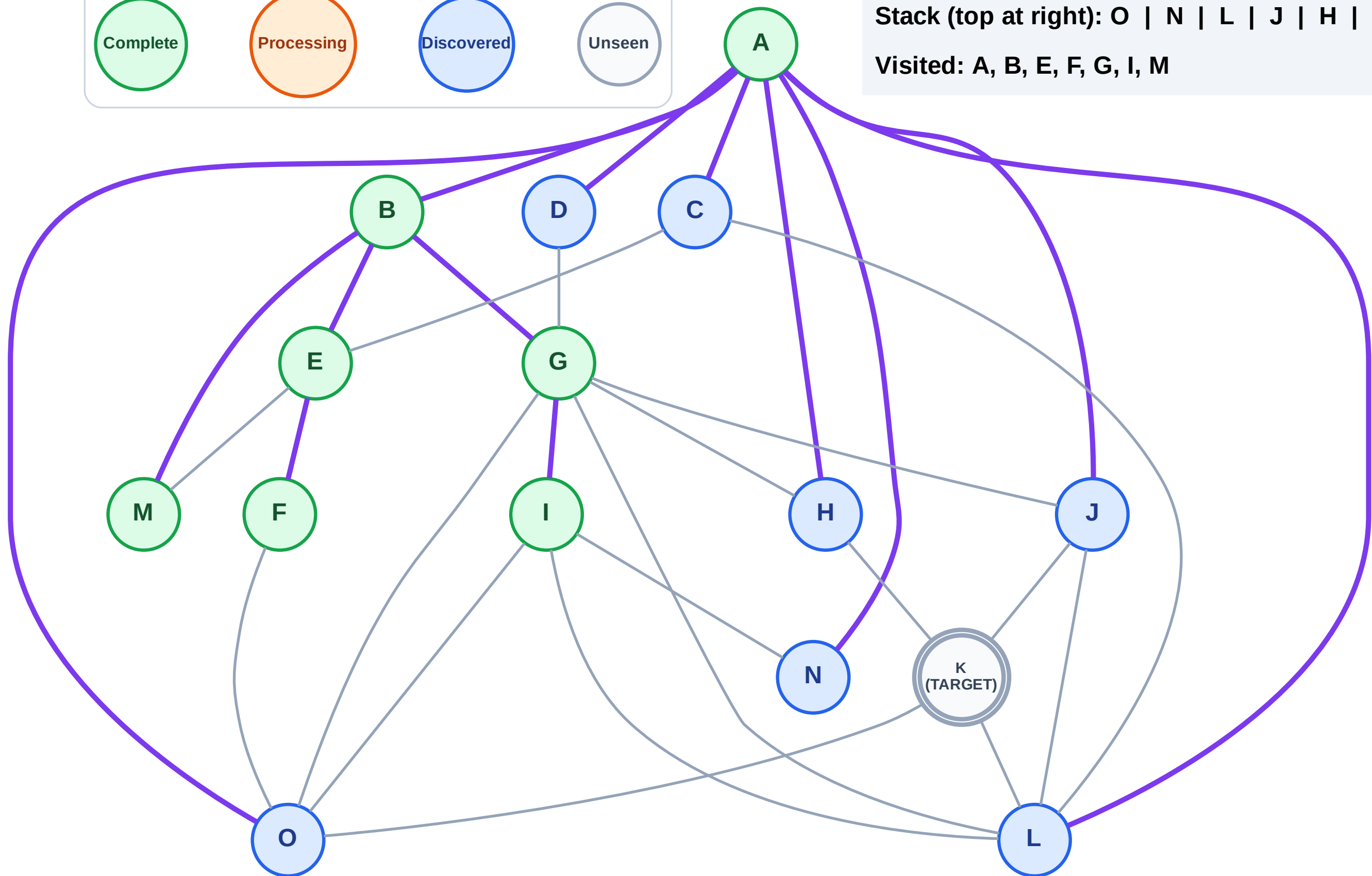
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C

Visited: A, B, E, F, G, I, M



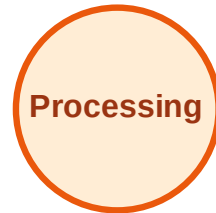
DFS Traversal

Step 16: Process node C

Legend



Complete



Processing



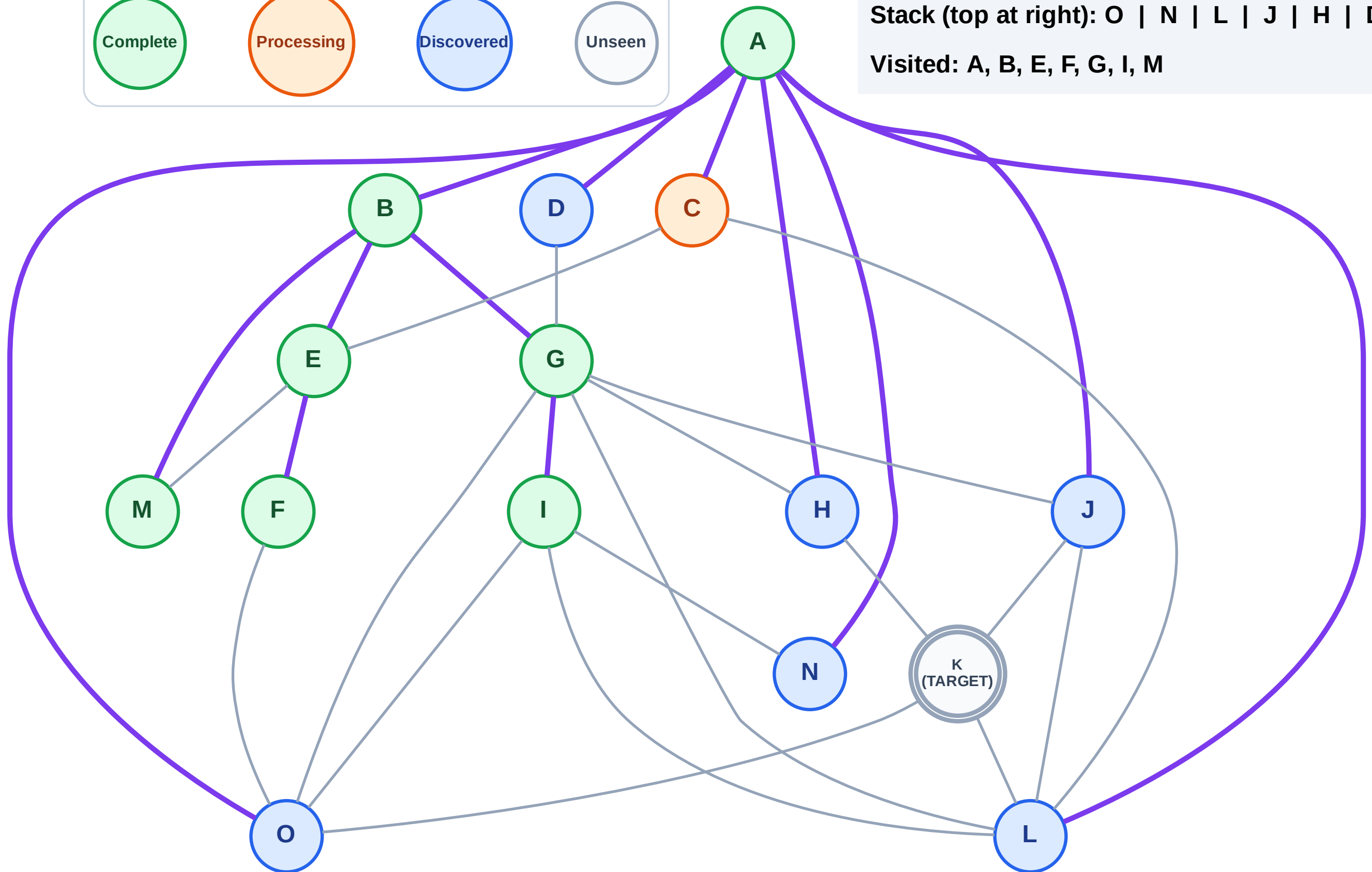
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D

Visited: A, B, E, F, G, I, M



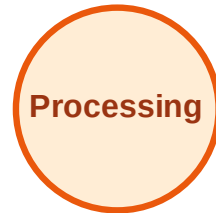
DFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



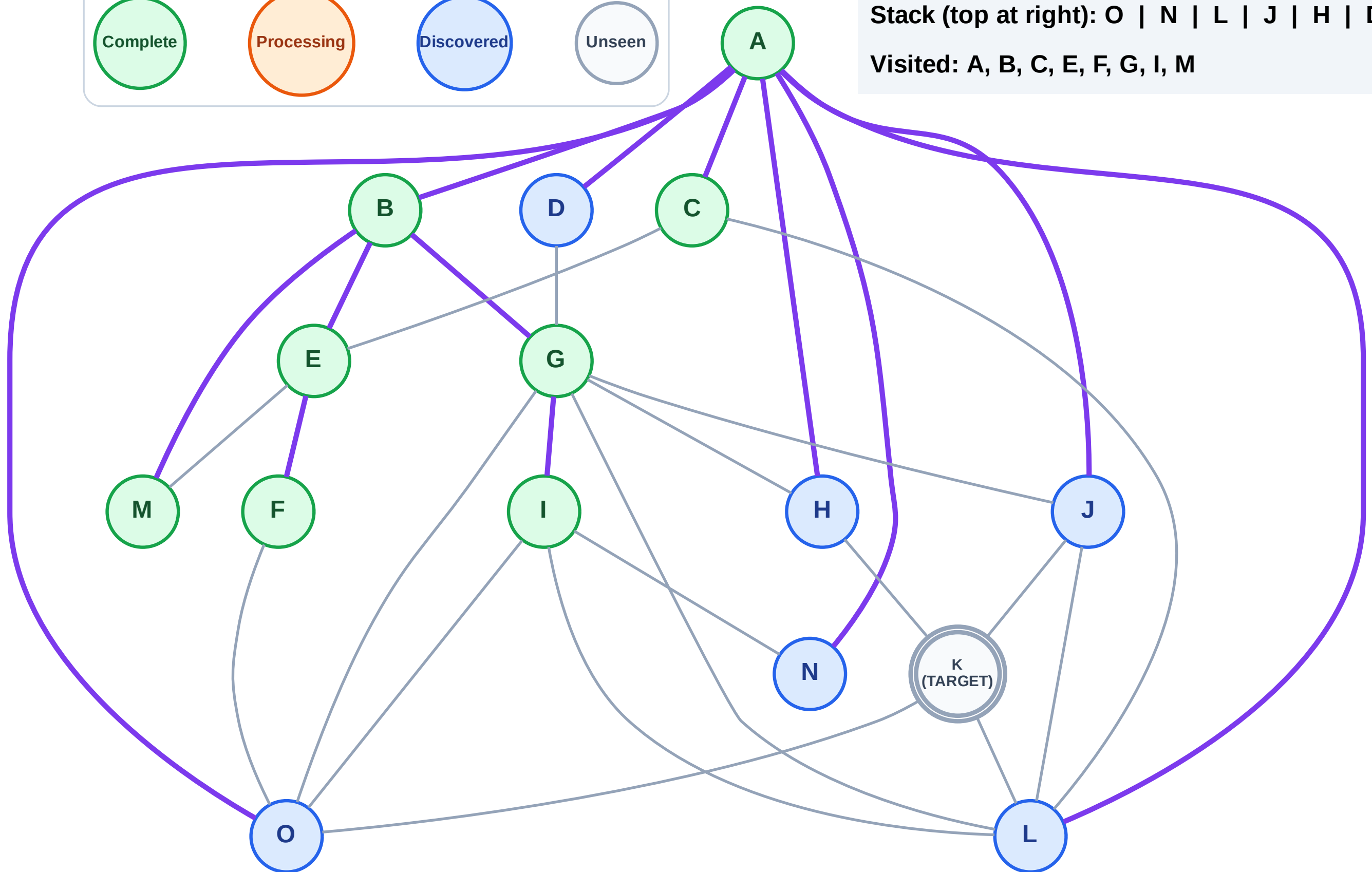
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D

Visited: A, B, C, E, F, G, I, M



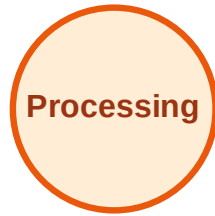
DFS Traversal

Step 18: Process node D

Legend



Complete



Processing



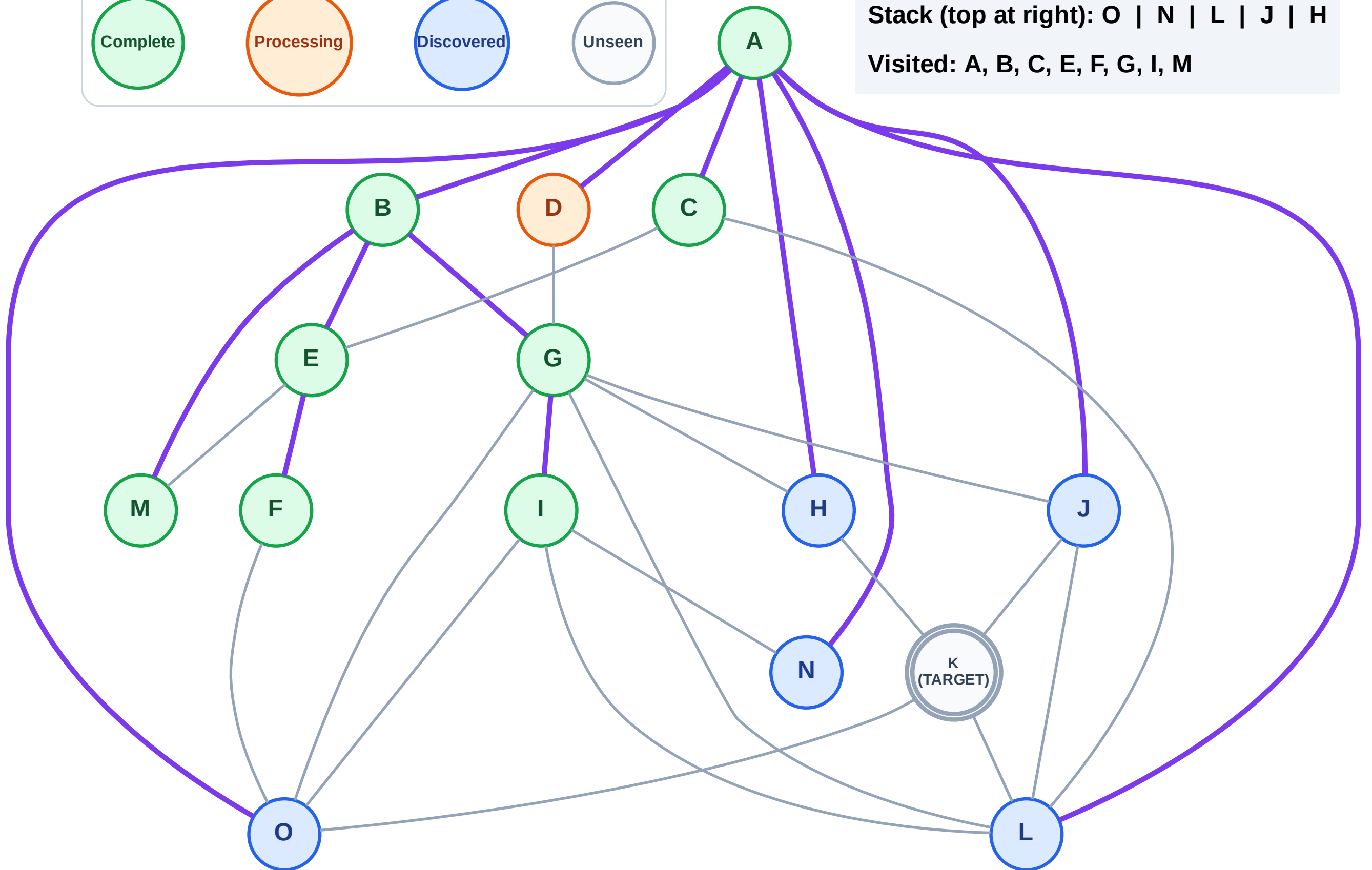
Discovered



Unseen

Stack (top at right): O | N | L | J | H

Visited: A, B, C, E, F, G, I, M

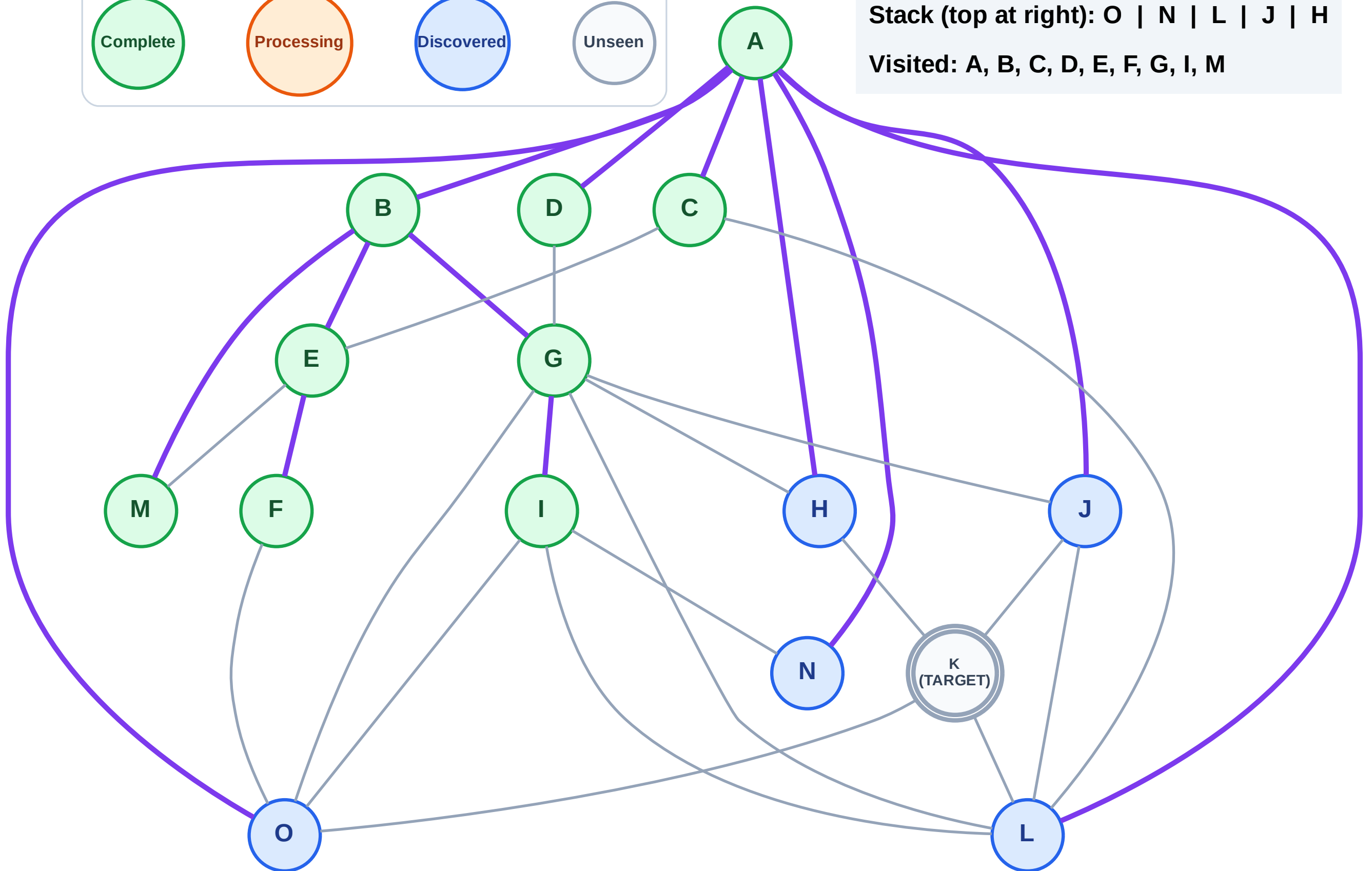


Step 19: No new neighbors from D

Legend



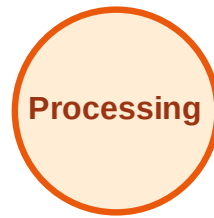
Visited: A, B, C, D, E, F, G, I, M



DFS Traversal

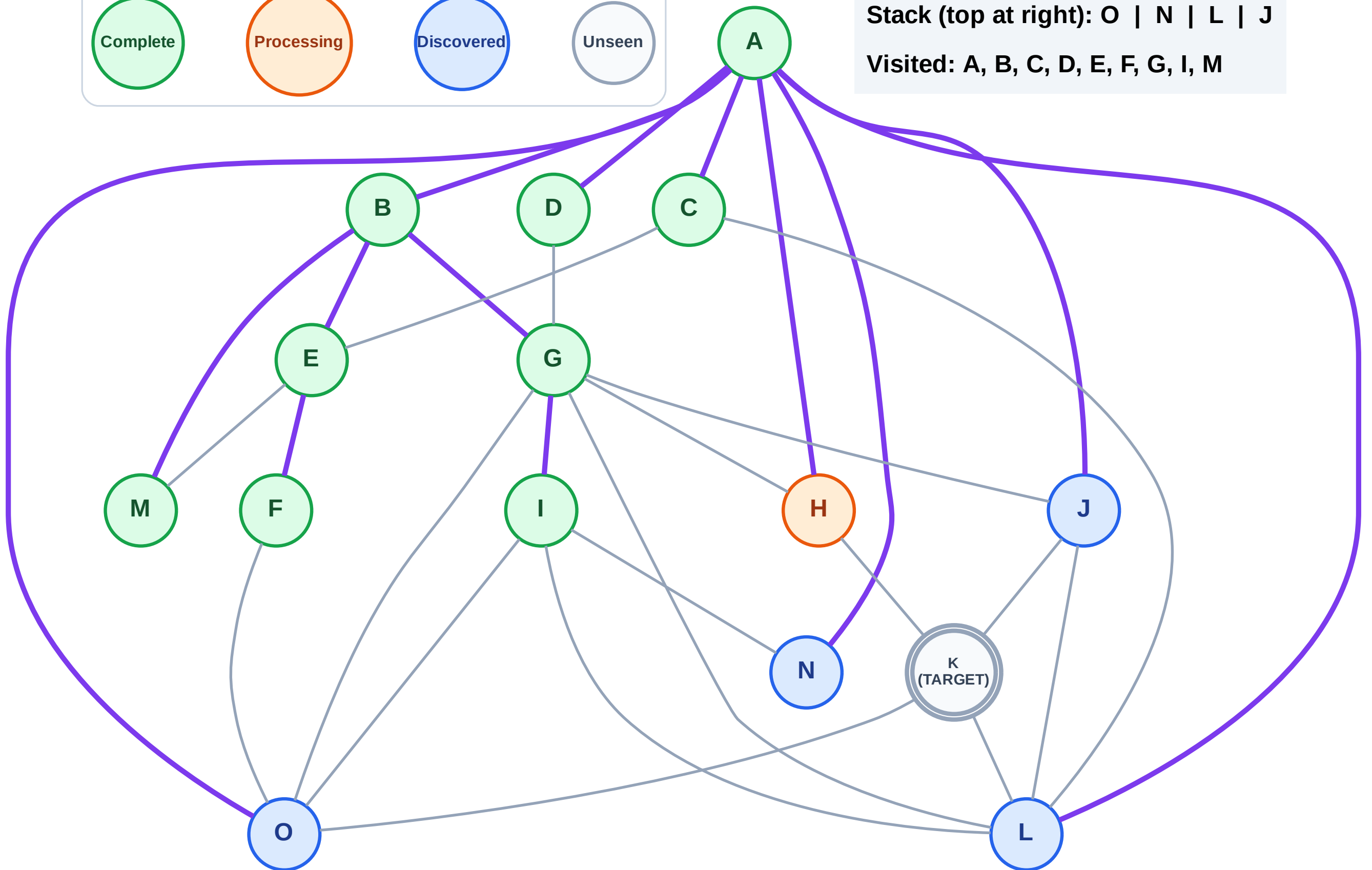
Step 20: Process node H

Legend



Stack (top at right): O | N | L | J

Visited: A, B, C, D, E, F, G, I, M

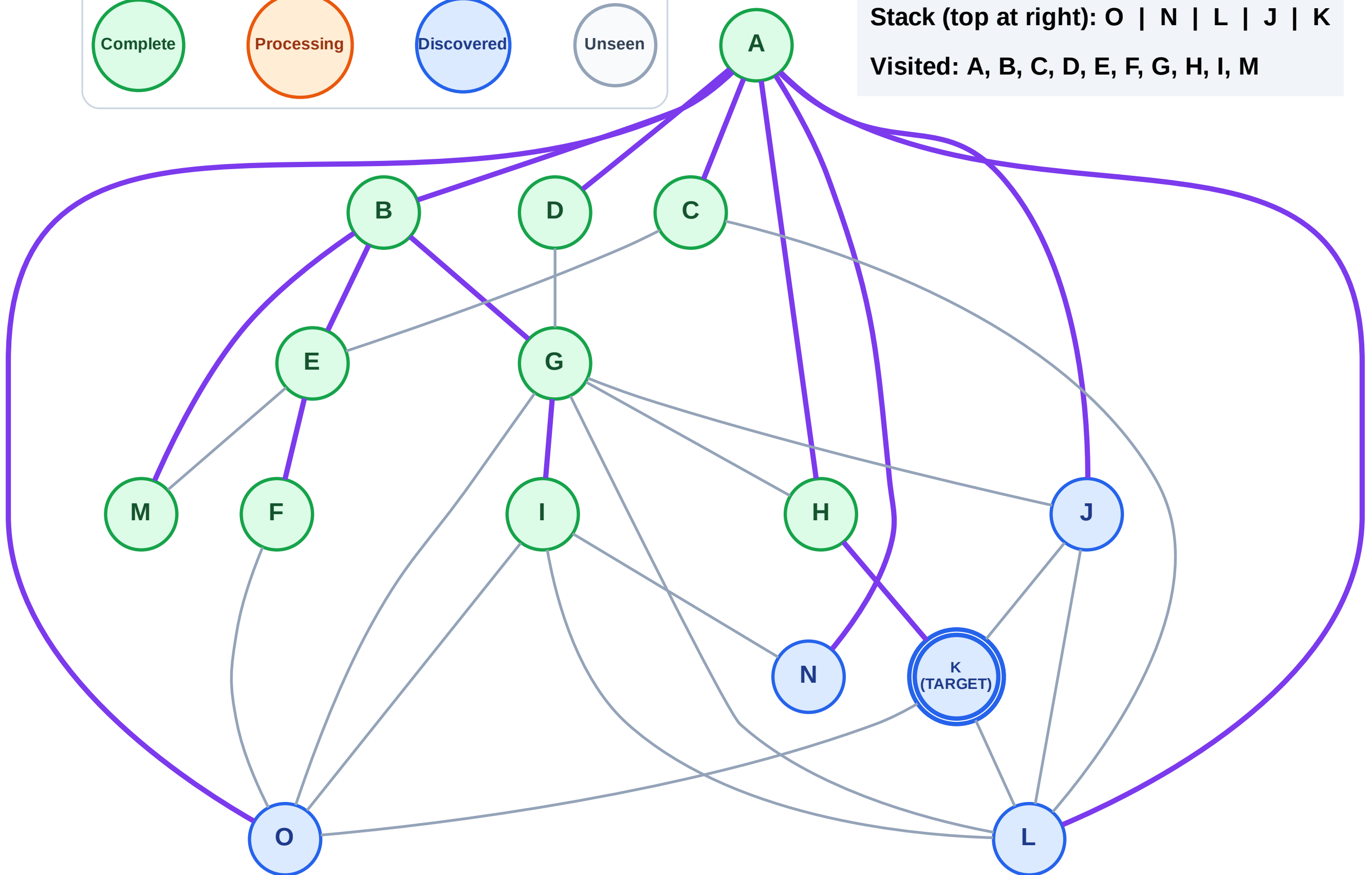


Step 21: Discover K from H

Legend

Unseen

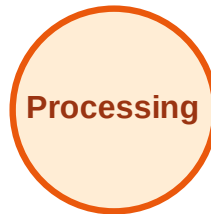
Visited: A, B, C, D, E, F, G, H, I, M



DFS Traversal

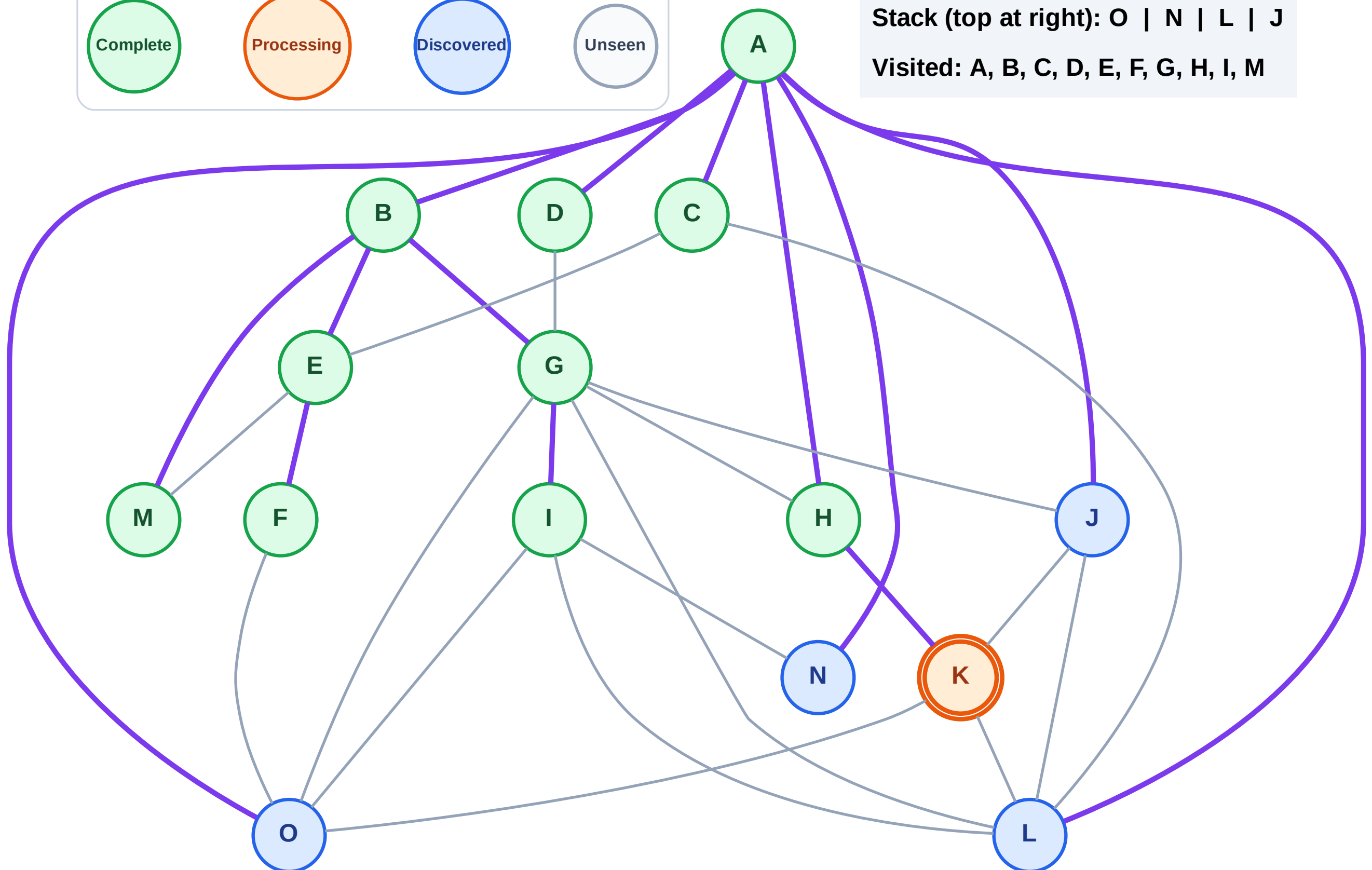
Step 22: Process node K

Legend



Stack (top at right): O | N | L | J

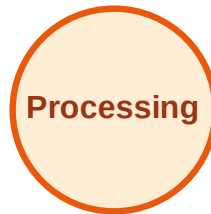
Visited: A, B, C, D, E, F, G, H, I, M



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → H → K

Legend



Stack (top at right): O | N | L | J
Visited: A, B, C, D, E, F, G, H, I, K, M

